



Bachelor in Artificial Intelligence

An AI-Based Decision Support System for Retinal OCT Classification: Comparing Deep Learning and Foundation Models

Supervisor:

Giovanna Nicora

Candidate:

Susanna Mazzocchi

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Susanna Mazzocchi

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A mia mamma e a mio papà, per il sostegno di sempre.

*A mia sorella, che mi ha aiutata più di quanto possa dire
e non ha mai smesso di crederci.*

*A me stessa, per non aver mai perso la motivazione
e non essermi mai fermata.*

Abstract

Retinal diseases such as Age-related Macular Degeneration (AMD) and Diabetic Macular Edema (DME) are among the leading causes of irreversible vision loss worldwide. Optical Coherence Tomography (OCT) has become the standard imaging modality for their diagnosis, yet the growing volume of scans generated daily far exceeds the capacity of available specialists.

This thesis presents ORCA, an AI-based decision support system for the automated classification of retinal OCT images into three diagnostic categories: AMD, DME, and Normal. Rather than focusing on a single architecture, this work systematically compares four modelling approaches spanning the current landscape of medical image classification: a lightweight custom convolutional neural network serving as a baseline, EfficientNet-B0 adapted via transfer learning from ImageNet, and two recent medical vision foundation models, MedGemma-4B and MIRAGE-Base. All models are trained and evaluated on the OCT5k dataset under a strict patient-level data splitting strategy that prevents data leakage by ensuring that all scans from the same individual are confined to a single partition.

Beyond classification accuracy, the system incorporates three complementary mechanisms for safe clinical deployment. Post-hoc calibration via temperature scaling aligns predicted confidence scores with empirical accuracy. Out-of-distribution detection through entropy scoring enables the system to flag inputs belonging to unseen pathologies, such as glaucoma, rather than silently assigning them to a known class. Visual explanations are provided through Gradient-weighted Class Activation Mapping for the convolutional models and occlusion sensitivity for the Vision Transformer backbone of MIRAGE-Base, allowing clinicians to verify that model decisions are grounded in anatomically plausible retinal structures.

The results demonstrate that classification performance, calibration, and out-of-distribution safety are partially independent dimensions that must be evaluated separately, and that no single model dominates across all of them. These findings offer practical guidance for selecting and deploying AI models in clinical ophthalmology, where accuracy alone is insufficient for trustworthy deployment.

Abstract (Italiano)

Le patologie retiniche, come la Degenerazione Maculare Legata all'Et  (Age-related Macular Degeneration, AMD) e l'Edema Maculare Diabetico (Diabetic Macular Edema, DME), rappresentano tra le principali cause di perdita irreversibile della vista a livello mondiale. La Tomografia a Coerenza Ottica (Optical Coherence Tomography, OCT)   diventata la metodica di imaging di riferimento per la loro diagnosi; tuttavia, il crescente numero di scansioni prodotte quotidianamente supera la capacit  di valutazione degli specialisti disponibili.

Questa tesi presenta ORCA, un sistema di supporto alle decisioni basato sull'intelligenza artificiale per la classificazione automatica di immagini OCT della retina in tre categorie diagnostiche: AMD, DME e retina normale. Piuttosto che concentrarsi su una singola architettura, il lavoro confronta sistematicamente quattro approcci rappresentativi dell'attuale panorama della classificazione di immagini mediche: una rete neurale convoluzionale leggera sviluppata come modello di riferimento, EfficientNet-B0 adattata mediante transfer learning da ImageNet e due recenti foundation models per la visione artificiale in ambito medico, MedGemma-4B e MIRAGE-Base. Tutti i modelli sono stati addestrati e valutati sul dataset OCT5k adottando una rigorosa suddivisione dei dati a livello di paziente, in modo da evitare fenomeni di data leakage garantendo che tutte le scansioni appartenenti allo stesso individuo fossero assegnate a un'unica partizione.

Oltre alle prestazioni di classificazione, il sistema integra tre meccanismi complementari finalizzati a un impiego clinico affidabile. La calibrazione post-hoc mediante *temperature scaling* allinea i livelli di confidenza predetti con l'accuratezza empirica. Il rilevamento di dati fuori distribuzione (*out-of-distribution detection*) tramite entropia consente al sistema di identificare immagini appartenenti a patologie non osservate durante l'addestramento, come il glaucoma, invece di assegnarle automaticamente a una delle classi note. Inoltre, vengono fornite spiegazioni visive delle decisioni del modello mediante Gradient-weighted Class Activation Mapping per i modelli convoluzionali e analisi di sensitivit  per occlusione (*occlusion sensitivity*) per il backbone Vision Transformer di MIRAGE-Base.

I risultati dimostrano che accuratezza di classificazione, calibrazione e robustezza rispetto a dati fuori distribuzione rappresentano dimensioni solo parzialmente correlate e devono pertanto essere valutate separatamente. Nessun modello risulta infatti superiore in tutti gli aspetti considerati. Questi risultati forniscono indicazioni pratiche per la selezione e l'adozione di modelli di intelligenza artificiale in oftalmologia clinica, dove la sola accuratezza non   sufficiente a garantirne un impiego affidabile.

Contents

1	Introduction to Retinal OCT Imaging	10
1.1	Retinal Anatomy and OCT Imaging	10
1.2	Motivation and Objectives	11
2	AI for Medical Image Processing: from Deep Learning to Foundation Models	13
2.1	Deep Learning for Medical Image Analysis	13
2.2	Vision Foundation Models in Medicine	15
2.3	Out-of-Distribution Detection in Medical AI	16
3	Dataset and Preprocessing	19
3.1	Dataset Description and Image Filtering	19
3.2	Preprocessing Pipeline	20
3.3	Data Augmentation	21
3.4	Train / Validation / Test Split	22
4	Models and Experimental Setup	25
4.1	Experimental Setup and Reproducibility	25
4.2	CNN Baseline and EfficientNet-B0	25
4.3	MedGemma-4B	26
4.4	MIRAGE	27
4.5	Uncertainty Quantification	28
5	Results and Evaluation	30
5.1	Classification Performance and Calibration	30
5.2	Out-of-Distribution Detection	36
5.3	Visual Explanations	40
5.4	Comparative Discussion	43
6	Prototype Development, Regulatory Framework, and Ethics	45
6.1	The ORCA Prototype	45
6.2	Regulatory Classification: ORCA as a SaMD	46
6.3	Trustworthiness Evaluation via the FUTURE-AI Framework	47
6.4	Ethical Tensions in Clinical Deployment	48
6.5	Pathway toward Clinical Deployment	49

7	Discussion and Conclusion	50
7.1	Summary of Contributions	50
7.2	Deep Learning vs Foundation Models: Comparative Analysis	50
7.3	Limitations	51
7.4	Future Directions	51
7.5	Closing Remarks	52
	References	53

List of Figures

1.1	Representative OCT cross-sections of the three diagnostic categories	11
2.1	Typical Convolutional Neural Network architecture	14
3.1	Visual representation of the dataset composition	20
3.2	Preprocessing pipeline	21
3.3	Class distribution across the three dataset splits	24
5.1	Normalised confusion matrices for all four models	33
5.2	One-vs-rest ROC curves for all four models	34
5.3	Optimal temperature T values for each model	35
5.4	Reliability diagrams (calibration curves) for all four evaluated models	36
5.5	Shannon entropy distributions after temperature scaling	39
5.6	OOD detection ROC curves using Shannon entropy	39
5.7	Grad-CAM saliency maps for test samples	40
5.8	Grad-CAM saliency maps for glaucoma OOD images	41
5.9	Occlusion sensitivity maps for MIRAGE-Base	42

List of Tables

3.1	Composition of the OCT5k dataset	20
3.2	Augmentation operations applied to the training set	22
3.3	Patient-level dataset split	23
5.1	Summary of classification performance	30
5.2	Per-class metrics for CustomCNN	31
5.3	Per-class metrics for EfficientNet-B0	31
5.4	Per-class metrics for MedGemma-4B	31
5.5	Per-class metrics for MIRAGE-Base	32
5.6	OOD baseline behaviour before temperature scaling	37
5.7	OOD detection performance after temperature scaling	37
6.1	System alignment matrix against FUTURE-AI framework	47

Chapter 1

Introduction to Retinal OCT Imaging

1.1 Retinal Anatomy and OCT Imaging

Retinal Anatomy

The retina is a thin layer of neural tissue lining the inner surface of the eye, responsible for converting light into electrical signals that are transmitted to the brain via the optic nerve. It is composed of several distinct layers, including the photoreceptor layer (rods and cones), the inner and outer nuclear layers, the ganglion cell layer, and the retinal pigment epithelium (RPE). The macula, a small central region of the retina, is responsible for high-acuity and colour vision. Damage to macular tissue, whether caused by age-related degeneration or metabolic disorders such as diabetes, leads to progressive and often irreversible vision loss.

OCT Imaging Principles and Acquisition

Optical Coherence Tomography (OCT) is a non-invasive imaging technique first introduced by Huang et al. [1] that produces high-resolution, cross-sectional images of biological tissue. OCT operates on the principle of low-coherence interferometry: a beam of near-infrared light is directed at the tissue, and the time delay of backscattered light is measured to reconstruct a two-dimensional image of internal microstructure. This process is analogous to ultrasound imaging, but uses light rather than sound, achieving axial resolutions on the order of a few micrometers [1]. In clinical ophthalmology, OCT has become the standard imaging modality for retinal examination, allowing clinicians to visualise individual retinal layers in vivo with a level of detail that was previously impossible through conventional examination alone [2]. A typical OCT acquisition consists of a series of A-scans, each representing a one-dimensional depth profile of optical reflectivity along a single axial line. A collection of adjacent A-scans forms a B-scan, which is the familiar cross-sectional image used in clinical practice. Modern spectral-domain OCT (SD-OCT) systems can acquire thousands of A-scans per second, enabling rapid volumetric imaging of the retina. Approximately 30 million OCT scans are performed worldwide each year, making it one of the most commonly used diagnostic procedures in ophthalmology [2].

Retinal Diseases Visible in OCT

Two of the most prevalent and clinically significant retinal diseases visible in OCT are Age-related Macular Degeneration (AMD) and Diabetic Macular Edema (DME). AMD currently affects approxi-

mately 170 million people worldwide, with projections estimating this number will reach 288 million by 2040 [3]. In its advanced neovascular form, AMD is characterised by the growth of abnormal blood vessels beneath the retina, a process known as choroidal neovascularisation (CNV). In OCT images, CNV appears as a hyperreflective neovascular membrane accompanied by subretinal fluid accumulation, which manifests as a dark, optically empty space between the neurosensory retina and the RPE [2]. Early-stage AMD is instead characterised by the presence of drusen, which are lipid deposits that accumulate beneath the RPE and appear as small hyperreflective elevations in the OCT scan.

DME is a vision-threatening complication of diabetic retinopathy, occurring when fluid accumulates within the layers of the retina due to increased vascular permeability. In OCT imaging, DME is identified by retinal thickening and the presence of intraretinal fluid, visible as hyporeflective cystoid spaces distributed within the retinal layers [2]. Unlike subretinal fluid in CNV, intraretinal fluid disrupts the normal layered architecture of the retina from within. Anti-VEGF therapy has become the primary treatment for both neovascular AMD and DME, and OCT imaging is essential for monitoring treatment response by tracking changes in fluid volume over time [3].

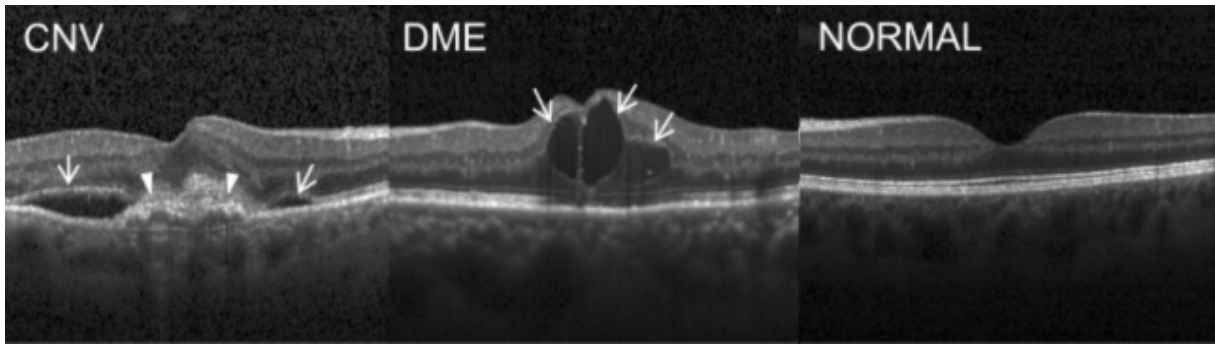


Figure 1.1: Representative OCT cross-sections of the three diagnostic categories. From left to right: choroidal neovascularisation (CNV) with subretinal fluid (arrows) and neovascular membrane (arrowheads); diabetic macular edema (DME) with intraretinal fluid (arrows); and a normal retina with preserved foveal contour. Adapted from [2].

The ability of OCT to clearly distinguish these pathological features makes it particularly well-suited for automated image classification. Normal retinas display a clean, well-defined layered structure with a preserved foveal contour and no signs of fluid or drusen, providing a clear visual contrast with pathological cases [2]. This characteristic visual separability between classes is one of the key properties that makes OCT a strong candidate for deep learning-based diagnostic systems, as discussed in the following chapter.

1.2 Motivation and Objectives

As the global incidence of retinal diseases rises, ophthalmology clinics face a growing logistical challenge. Modern OCT systems generate an overwhelming volume of high-resolution scans every day, yet the number of specialists available to review them remains small. This imbalance creates a severe bottleneck in patient care. Relying strictly on manual interpretation is no longer sustainable; it is inherently time-consuming, prone to inter-observer variability, and fundamentally difficult to scale to the demands of modern practice [2]. Deep learning-based decision support systems offer a practical solution to this crisis, promising fast and reproducible diagnostic assistance.

Motivated by these clinical demands, this thesis develops and evaluates an automated pipeline designed to classify OCT cross-sections into three primary diagnostic categories: AMD, DME, and Normal. Rather than focusing on a single architecture, this work systematically compares four distinct modelling approaches. A standard convolutional neural network (CNN) is established as a baseline alongside an EfficientNet-B0 model leveraging transfer learning. These are then compared against two recent medical foundation models, MedGemma-4B and MIRAGE-Base, to see how newer architectures perform.

Importantly, raw accuracy alone is not enough for a model to be actually trusted in clinical practice. To bridge the gap between theoretical performance and trustworthy clinical deployment, the proposed system explicitly incorporates out-of-distribution (OOD) detection and visual explainability mechanisms. Ultimately, the objective is not simply to automate a classification task, but to engineer a robust diagnostic tool that clinicians can interpret, verify, and safely rely on in their daily workflow.

Chapter 2

AI for Medical Image Processing: from Deep Learning to Foundation Models

2.1 Deep Learning for Medical Image Analysis

Convolutional Neural Networks

Deep learning refers to a class of machine learning methods based on artificial neural networks with multiple layers of processing units. These models learn hierarchical representations of data directly from raw input, eliminating the need for manual feature engineering. In the context of image classification, Convolutional Neural Networks (CNNs) have become the dominant architecture since the seminal work of [4], which demonstrated that networks trained end-to-end with gradient-based optimisation could outperform hand-crafted feature extraction methods on complex visual recognition tasks.

The core building block of a CNN is the convolutional layer, which applies a set of learnable filters across the spatial dimensions of the input. Each filter produces a feature map that captures a specific local pattern, such as edges, textures, or shapes. This operation exploits two key properties: local connectivity, meaning each neuron responds only to a small region of the input, and weight sharing, meaning the same filter is applied across the entire input. These properties make CNNs significantly more parameter-efficient than fully connected networks when processing images [4]. Convolutional layers are typically followed by a non-linear activation function. The most widely used is the Rectified Linear Unit (ReLU), defined as:

$$f(x) = \max(0, x)$$

This function sets all negative activations to zero, introducing the non-linearity necessary for the network to learn complex decision boundaries.

Pooling layers are interleaved between convolutional layers to progressively reduce the spatial resolution of feature maps, thereby decreasing computational cost and introducing a degree of translation invariance. Max pooling, the most commonly used variant, retains only the maximum activation within each pooling region. After several convolutional and pooling blocks, the feature maps are flattened and passed through one or more fully connected layers that produce the final class scores.

Training and Optimisation

Training a CNN requires minimising a loss function that measures the discrepancy between the model's predictions and the ground truth labels. For multi-class classification, the standard choice is cross-entropy loss, computed over the softmax output of the network. Minimisation is performed via back-propagation, which computes the gradient of the loss with respect to every parameter in the network using the chain rule of calculus. These gradients are then used to update the parameters through an optimisation algorithm such as Stochastic Gradient Descent (SGD) or Adam, the latter of which adapts the learning rate for each parameter based on estimates of first and second order moments of the gradients.

Batch Normalisation [5] is a widely adopted technique that normalises the activations of each layer across a mini-batch during training. This accelerates convergence by reducing internal covariate shift and allows the use of higher learning rates. It has become a standard component in modern CNN architectures.

Regularisation and Data Augmentation

Regularisation techniques are essential to prevent overfitting, particularly when training on limited data. Dropout [6] randomly sets a fraction of neuron activations to zero during each training step, forcing the network to learn redundant representations and reducing co-adaptation between neurons. Weight decay, also known as L_2 regularisation, adds a penalty proportional to the squared magnitude of the weights to the loss function, discouraging excessively large parameter values. Label smoothing is a further technique that replaces hard one-hot target labels with soft targets, slightly reducing overconfidence in the model's predictions and improving generalisation.

Data augmentation artificially increases the effective size of the training set by applying random transformations to input images at each training step. Common augmentations for medical images include random horizontal flipping, small rotations, colour jitter, and Gaussian blur [7]. These transformations encourage the model to learn features that are invariant to irrelevant variations in the input, which is particularly valuable in settings where labelled data is scarce, as is typical in medical imaging applications.

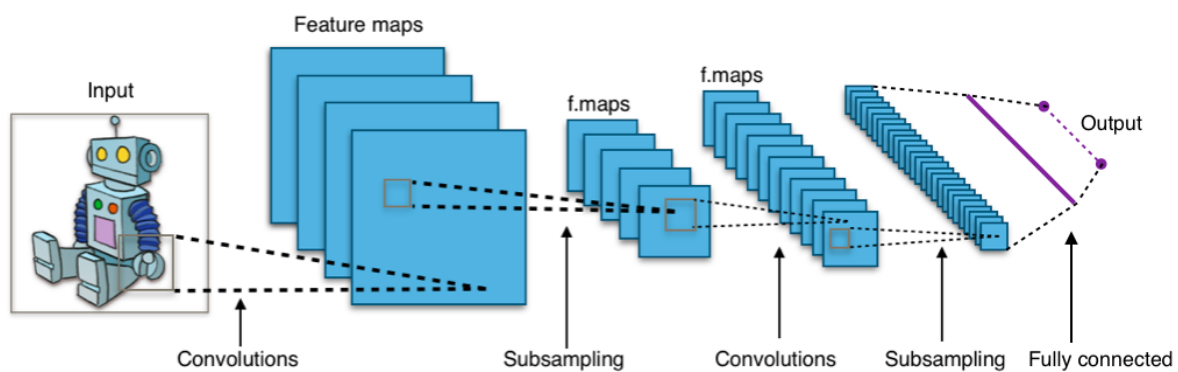


Figure 2.1: Typical Convolutional Neural Network architecture showing alternating convolutional and pooling layers for feature extraction, followed by fully connected layers for classification. Adapted from [4].

Transfer Learning and Fine-Tuning

While CNNs trained from scratch can achieve strong performance on large datasets, training on limited labelled data, as is typical in medical imaging, requires a different approach. Transfer learning addresses this challenge by initialising a model with weights pre-trained on a large general-purpose dataset, and then adapting it to a specific target task [8]. The most widely used source of pre-trained weights is ImageNet, a benchmark dataset containing over 1.2 million images across 1000 categories [9]. Networks pre-trained on ImageNet develop rich hierarchical representations in their early layers, such as edge detectors and texture filters, that generalise well across many visual domains, including medical imaging [10].

The standard transfer learning procedure consists of replacing the final classification layer of a pre-trained network with a new head suited to the target number of classes, and then fine-tuning the model on the target dataset. Two main strategies are commonly adopted. In the *frozen backbone* approach, the pre-trained convolutional layers are kept fixed and only the new classification head is trained. This is computationally efficient and works well when the target dataset is small or when it is sufficiently similar to the pre-training domain. In contrast, *end-to-end fine-tuning* updates all layers of the network jointly, allowing the representations to adapt more specifically to the target task. This typically yields better performance but requires more data and careful regularisation to avoid overfitting.

The EfficientNet Architecture

EfficientNet [11] is a family of convolutional neural networks designed around a principled compound scaling method. Rather than scaling only one dimension of the network as was common in prior work, such as increasing only depth as in ResNet [12] or only width, the authors propose to uniformly scale network depth, width, and input resolution together using a fixed compound coefficient ϕ . Formally, the scaling rules are defined as:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi$$

where α , β , and γ are constants determined by a small grid search, subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. This constraint ensures that the total computational cost scales approximately as 2^ϕ FLOPs per doubling of resources. The baseline architecture, EfficientNet-B0, is obtained via neural architecture search and uses Mobile Inverted Bottleneck Convolution (MBConv) blocks [11]. Larger variants EfficientNet-B1 through B7 are obtained by progressively increasing ϕ .

2.2 Vision Foundation Models in Medicine

From Task-Specific Models to Foundation Models

Foundation models are large neural networks trained on broad, diverse datasets that develop general-purpose representations transferable to a wide range of downstream tasks. Unlike task-specific models trained from scratch, foundation models are designed to be adapted with minimal additional training data, making them particularly valuable in domains such as medical imaging where labelled data is limited [13]. In the medical vision domain, several foundation models have been proposed in recent years, each targeting different clinical modalities and tasks. This work employs two of them: MedGemma and MIRAGE.

MedGemma

MedGemma is an open collection of medical vision-language foundation models developed by Google Research and Google DeepMind [13]. It is built upon the Gemma 3 architecture and is available in two variants: a 4B multimodal model capable of processing both images and text, and a 27B text-only model. The 4B variant, used in this work, integrates a vision encoder with a large language model to support a broad range of medical image understanding tasks across radiology, ophthalmology, dermatology, and histopathology.

The visual understanding capabilities of MedGemma are powered by MedSigLIP, a medically fine-tuned version of the SigLIP vision encoder [13]. SigLIP is trained with a sigmoid contrastive loss rather than the standard softmax used in CLIP-style models, which improves performance at smaller batch sizes and scales more efficiently. MedSigLIP is obtained by continuing the training of SigLIP-400M on a curated collection of medical images spanning multiple modalities, including retinal fundus images and chest X-rays. The encoder processes images at a native resolution of 896×896 pixels, producing a sequence of spatial tokens that capture fine-grained visual features relevant to clinical diagnosis.

MIRAGE

MIRAGE (Multimodal foundation model and benchmark for comprehensive retinal OCT image analysis and segmentation) is a multimodal foundation model for the analysis of retinal OCT and scanning laser ophthalmoscopy (SLO) images, developed at the Medical University of Vienna [14]. Unlike MedGemma, which is a general-purpose medical model adapted to multiple clinical domains, MIRAGE is specifically designed for retinal imaging and exploits the complementary information between OCT and SLO modalities during pretraining.

MIRAGE is based on a Vision Transformer (ViT) encoder and is pretrained on VIBES, a large in-house dataset of 261,184 paired OCT-SLO samples from 42,082 patients, using a multimodal Masked Autoencoder (MAE) approach [14]. The pretraining objective consists of reconstructing all input modalities from masked versions of the same images, which forces the model to learn the complementary relationships between OCT and SLO. Retinal layer pseudo-labels, generated automatically, are included as a third modality to further enrich the learned representations. This domain-specific multimodal pretraining strategy distinguishes MIRAGE from other ophthalmic foundation models such as RETFound, which is trained on a single modality, and EyeCLIP, which mixes unpaired multimodal data without exploiting fine-grained inter-modal relationships [14].

MIRAGE has been evaluated on a comprehensive benchmark of 19 tasks from 14 datasets, covering OCT and SLO classification and segmentation. It significantly outperforms state-of-the-art foundation models including DINOv2 and RETFound across all task types, with an average AUROC of 95.59% on OCT classification tasks [14]. These results motivate its adoption in this work as a strong baseline for OCT-based retinal disease classification.

2.3 Out-of-Distribution Detection in Medical AI

A well-known limitation of deep neural networks is that they tend to produce high-confidence predictions even when presented with inputs that fall outside their training distribution [15]. In standard classification, the softmax function assigns a probability to each class and the model predicts the class with the highest value. Because the softmax is computed using the exponential function, even small

differences in the logits can produce strongly peaked output distributions. As a result, a model trained on a fixed set of categories may assign very high confidence to an entirely unseen input, rather than expressing uncertainty. This is illustrated concretely by Hendrycks and Gimpel [15], who show that random Gaussian noise fed into an image classifier yields a predicted class probability of 91%, highlighting how misleading raw softmax probabilities can be as direct confidence estimates.

In a medical context, this behaviour is particularly concerning. A diagnostic system that silently assigns a high-confidence label to an unseen pathology gives no signal to the clinician that something unusual has occurred. In the context of this work, the models are trained on three OCT categories: AMD, DME, and Normal retinas. Glaucoma, which produces structurally distinct OCT patterns, is not among the training classes. When a glaucoma image is presented at test time, the model will inevitably assign it to one of the known categories. OOD detection aims to identify such inputs and flag them for human review before a potentially harmful decision is propagated.

Confidence-Based Detection and Entropy Scoring

A simple but effective baseline for detecting misclassified and out-of-distribution examples was proposed by Hendrycks and Gimpel [15], using the maximum softmax probability as a detection score. The central observation is that, while the absolute value of the predicted class probability is not a reliable indicator of confidence, correctly classified in-distribution examples tend to have higher maximum softmax probabilities than OOD ones. Given a classifier with softmax output $\hat{p}(x)$, the detection score is defined as:

$$s(x) = \max_k \hat{p}_k(x) \quad (2.1)$$

A threshold τ is then applied so that inputs with $s(x) < \tau$ are flagged as potentially OOD. Despite its simplicity, this baseline achieves strong performance across a range of tasks in computer vision, natural language processing, and speech recognition [15].

A closely related approach is entropy-based scoring. The predictive entropy of the softmax distribution is:

$$H(\hat{p}) = - \sum_k \hat{p}_k(x) \log \hat{p}_k(x) \quad (2.2)$$

In-distribution inputs tend to produce confident, peaked distributions with low entropy, while OOD inputs tend to yield flatter distributions and therefore higher entropy. Entropy-based scoring is closely related to the KL divergence of the softmax distribution from a uniform distribution, and the two approaches yield highly correlated detection performance in practice [15].

Temperature Scaling and Calibration

A practical limitation of raw softmax probabilities is that modern deep neural networks are often overconfident: their predicted class probability tends to be systematically higher than their empirical accuracy [16]. Temperature scaling is a simple post-hoc calibration method that addresses this by dividing the logits by a scalar parameter $T > 1$ before applying the softmax:

$$\hat{p}_k^T(x) = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)} \quad (2.3)$$

A larger temperature produces a softer probability distribution, bringing confidence estimates into better alignment with actual accuracy. The parameter T is optimised on a held-out validation set by

minimising the negative log-likelihood. Because temperature scaling does not alter the argmax prediction, classification accuracy is perfectly preserved while the reliability of confidence scores improves substantially [16].

Combining calibrated confidence estimates with a detection threshold enables selective prediction: rather than always returning a hard classification, the system can abstain when its confidence falls below a given value [15]. In a clinical deployment scenario, this abstention mechanism acts as a safety layer, routing uncertain or anomalous inputs to a human expert rather than producing an unreliable automated output.

Explainability and Human Oversight

Beyond uncertainty quantification, trustworthy clinical AI requires that model decisions be interpretable by the clinician. The machine learning literature has shown that a model’s expressed confidence can be highly misleading, particularly when facing inputs that differ from the training distribution [17]. Explainability methods address this gap by making the model’s internal reasoning accessible to clinical users, complementing uncertainty estimates with spatial evidence of where the model is looking.

Gradient-weighted Class Activation Mapping (Grad-CAM), proposed by [18], produces a class-discriminative visual explanation by computing the gradient of the predicted class score with respect to the final convolutional feature maps, globally average-pooling them to obtain neuron importance weights α_k^c , and combining them as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \quad (2.4)$$

The resulting heatmap highlights the image regions that most influenced the prediction for the target class. In a clinical setting, this allows a clinician to verify whether the model attended to anatomically plausible structures, providing the spatial grounding needed for meaningful human oversight.

Chapter 3

Dataset and Preprocessing

3.1 Dataset Description and Image Filtering

Source and Structure

The dataset used in this work is OCT5k [19], a publicly available multi-disease retinal OCT dataset developed at University College London (UCL) in collaboration with multiple international clinical centres. The dataset was originally compiled to support retinal layer segmentation and biomarker detection tasks, providing pixel-wise expert annotations for 1,672 OCT B-scans across three diagnostic categories: Age-related Macular Degeneration (AMD), Diabetic Macular Edema (DME), and healthy (Normal) subjects. While the original dataset primarily targets segmentation with this annotated subset, the associated raw image directories contain a much larger collection of scans from the same clinical cohort. Since our task is image-level classification rather than pixel-wise segmentation, we leverage this entire broader collection, making it equally suitable as a classification benchmark.

The images were acquired using Heidelberg Spectralis SD-OCT devices, which produce cross-sectional B-scans with a standard resolution of 496×512 pixels. Each image filename encodes the patient identifier directly, enabling unambiguous mapping between images and individual subjects. No demographic metadata such as age or sex is provided in the dataset. The data is organised into five sub-folders reflecting the original annotation batches: AMD Part1, AMD Part2, DME, Normal Part1, and Normal Part2. For the purposes of this work, the two AMD sub-folders and the two Normal sub-folders were merged into single class directories, yielding a three-class dataset.

Image Filtering and Quality Control

A complete scan of the dataset directories yielded 3,348 image files in total. Prior to any further processing, a resolution-based quality control step was applied to detect and remove non-OCT images present in the collection. Inspection revealed that 117 images deviated from the expected 496×512 resolution, instead exhibiting a 768×768 pixel format characteristic of colour fundus photography. These fundus images represent a fundamentally different imaging modality and would constitute modality contamination if included in a model trained exclusively on OCT B-scans. All 117 outlier images were therefore removed, leaving a clean dataset of **3,231 OCT images** across 117 patients. Table 3.1 summarises the composition of the filtered dataset.

Table 3.1: Composition of the OCT5k dataset after filtering non-OCT images.

Class	Patients	Images	Img / Patient	Share (%)
AMD	18	486	27.0	15.0
DME	49	1,117	22.8	34.6
Normal	50	1,628	32.6	50.4
Total	117	3,231	27.6	100.0

To visually contextualise these numbers, Figure 3.1 illustrates the distribution of the filtered dataset.

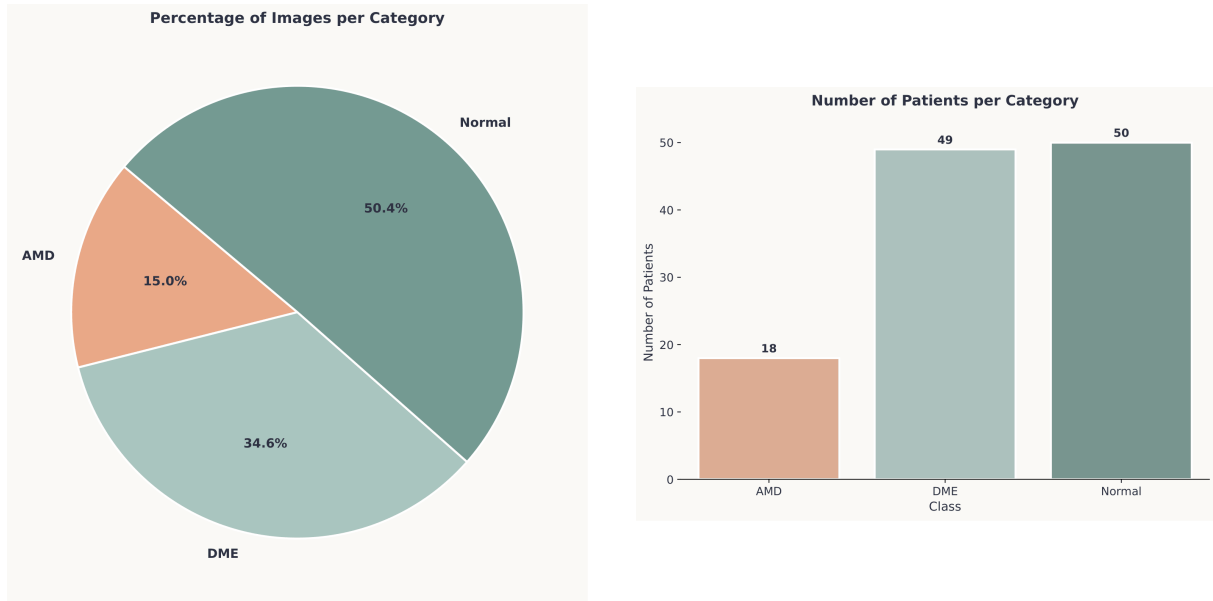


Figure 3.1: Visual representation of the dataset composition. The pie chart (a) highlights the class imbalance at the individual scan level, while the bar chart (b) demonstrates how this imbalance persists across the number of unique patients per clinical category.

The class distribution is moderately imbalanced, as evident in both Table 3.1 and Figure 3.1: Normal images account for approximately half of the dataset, while AMD images represent only 15.0%. This imbalance reflects the clinical reality of the source cohort and is addressed during model training through class-weighted loss functions. The number of images per patient varies across classes, ranging from approximately 23 in the DME group to over 32 in the Normal group, a variability that arises from differences in imaging protocols and the number of follow-up scans available per subject. Because multiple images originate from the same patient, a naive random split of the image set would risk placing images of the same patient in both the training and test sets, artificially inflating evaluation metrics. This data leakage risk is addressed below by the patient-level splitting strategy.

3.2 Preprocessing Pipeline

All images in the dataset undergo a fixed preprocessing pipeline before being fed to any model. The pipeline, illustrated in Figure 3.2, consists of six sequential steps applied uniformly across all splits, with the exception of the final normalisation step which is applied only to the training set as part of the augmentation strategy.

Raw images, stored in TIFF, PNG, or JPEG format, are first converted to grayscale using a single-channel representation. This choice is motivated by the nature of OCT imaging: B-scans are inherently monochromatic, and any colour information present in the original files is an artefact of the export format rather than clinically meaningful signal. Retaining a single channel also reduces memory consumption and computational cost without any loss of diagnostic information.

Images are then resized to a fixed resolution of 512×512 pixels using Lanczos resampling, which preserves edge sharpness better than bilinear or nearest-neighbour interpolation and is therefore preferable when fine structural details such as retinal layer boundaries are clinically relevant. The choice of 512×512 provides a resolution large enough to retain the characteristic features of each pathology while remaining computationally tractable for batch training on GPU.

Pixel values are subsequently scaled from the original integer range $[0, 255]$ to the continuous range $[0, 1]$ by dividing by 255.0 and casting to `float32`. Finally, the tensor is normalised to the range $[-1, 1]$ using `mean = 0.5` and `standard deviation = 0.5`, a standard transformation that centres the distribution around zero and produces inputs compatible with the expected statistics of ImageNet-pretrained models [7]. This normalisation step is applied uniformly across all splits, training, validation, and test, to ensure consistent input statistics at inference time.

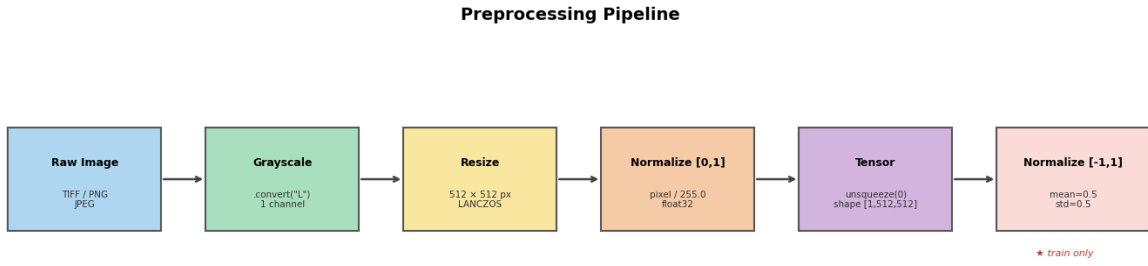


Figure 3.2: Sequential preprocessing steps applied to all OCT images before training. The final normalisation to $[-1, 1]$ (marked with ★) is applied to all splits.

3.3 Data Augmentation

Data augmentation is applied exclusively to the training set to artificially increase the variability of the observed samples and reduce the risk of overfitting. Validation and test images receive only the normalisation step described in Section 3.2, so that evaluation metrics reflect performance on unmodified inputs.

The augmentation pipeline is implemented as a `torchvision.transforms` composition and is passed to the `OCTDataset` class at instantiation time. It consists of five stochastic operations applied in the following order.

A **random horizontal flip** is applied with the default probability of 0.5. Horizontal mirroring is anatomically neutral for retinal B-scans because the pathological features of interest (drusen deposits in AMD, intraretinal and subretinal fluid in DME, and the intact layer structure in Normal scans) are symmetric with respect to the vertical axis and do not carry laterality information in cross-sectional OCT images.

A **random rotation** of up to $\pm 10^\circ$ introduces mild in-plane orientation variability. The small rotation range is deliberate: OCT B-scans are acquired along a fixed horizontal plane, so large rotations would produce unrealistic deformations of the retinal architecture.

A **random affine translation** of up to 5% of the image width and height in each axis shifts the retinal content within the frame. This simulates variability in the positioning of the scan window across patients and acquisition sessions without altering the morphology of the imaged tissue.

Brightness and contrast jitter (magnitude 0.3 for both parameters) is applied with probability $p = 0.5$. This operation accounts for inter-device and inter-session variability in signal intensity, which is a known source of domain shift in OCT data even when the same device model is used across clinical centres [7].

Finally, a **Gaussian blur** with kernel size 5×5 and σ uniformly sampled from $[0.1, 1.5]$ is applied with probability $p = 0.3$. Mild blurring reduces the model’s reliance on high-frequency texture artefacts introduced by the resampling step and encourages learning of coarser structural features such as layer boundaries and fluid pockets.

The normalisation to $[-1, 1]$ (mean = 0.5, std = 0.5) is appended at the end of the training pipeline and is identically present in the validation and test pipeline, ensuring consistent input statistics across all splits. Table 3.2 summarises the full augmentation configuration.

Table 3.2: Augmentation operations applied to the training set. Val/Test receives only the final normalisation step.

Operation	Parameters	Probability	Split
RandomHorizontalFlip	—	0.5	Train only
RandomRotation	$\pm 10^\circ$	1.0	Train only
RandomAffine	translate (5%, 5%)	1.0	Train only
ColorJitter	brightness = 0.3, contrast = 0.3	0.5	Train only
GaussianBlur	$k = 5$, $\sigma \in [0.1, 1.5]$	0.3	Train only
Normalize $[-1, 1]$	mean = 0.5, std = 0.5	1.0	All splits

3.4 Train / Validation / Test Split

Motivation for Patient-Level Splitting

As noted in Section 3.1, the OCT5k dataset contains multiple B-scans per patient, with an average of 27.6 images per subject. A naive image-level random split would therefore risk placing scans of the same patient in both the training and test partitions. Because scans from the same eye share the same underlying pathology, retinal morphology, and acquisition parameters, a model trained on some scans and evaluated on others from the same patient would face an artificially easy classification task: it could achieve high accuracy simply by memorising patient-specific visual features rather than learning class-discriminative representations. This form of data leakage would produce overly optimistic generalisation estimates and undermine the clinical validity of the evaluation. All splitting decisions are therefore made at the *patient level*, ensuring complete separation of patient identities across the three subsets.

Splitting Strategy

The split is implemented using `StratifiedGroupKFold` from `scikit-learn`, which simultaneously enforces group integrity (patients are never shared across folds) and stratification (the relative class frequencies of AMD, DME, and Normal are approximately preserved within each fold). A fixed random seed of 42 is used throughout to guarantee reproducibility.

The procedure unfolds in two stages. In the first stage, a `StratifiedGroupKFold` with `n_splits = 3` is applied to the full dataset. The first fold is extracted, yielding approximately $\frac{2}{3}$ of patients for training and the remaining $\frac{1}{3}$ as a temporary pool. In the second stage, a `StratifiedGroupKFold` with `n_splits = 2` is applied to the temporary pool, dividing it into equally sized validation and test subsets of approximately $\frac{1}{6}$ each. The resulting partition is therefore approximately 66%/18%/16% at the patient level, which translates to a comparable image-level split given the relatively uniform image-per-patient ratio across classes.

Split Composition

Table 3.3 reports the composition of the three subsets in terms of both unique patients and images. The training set contains 77 patients (2,093 images), the validation set 21 patients (585 images), and the test set 19 patients (553 images).

Table 3.3: Patient-level dataset split, with resulting image counts after stratified group assignment.

Subset	Patients	Images	Patients (%)	Images (%)
Training	77	2,093	65.8	64.8
Validation	21	585	17.9	18.1
Test	19	553	16.2	17.1
Total	117	3,231	100.0	100.0

The class distribution is preserved across splits by construction, as illustrated in Figure 3.3. Class weights for the training loss are computed after splitting using `sklearn.utils.class_weight.compute_class_weight` with the 'balanced' option, which assigns inverse-frequency weights ($w_c = \frac{N}{K \cdot n_c}$, where N is the total number of training images, K the number of classes, and n_c the number of training images in class c). The resulting weights are higher for AMD, the minority class, and lower for Normal, the majority class, thereby counteracting the imbalance at the loss level without modifying the data distribution itself.

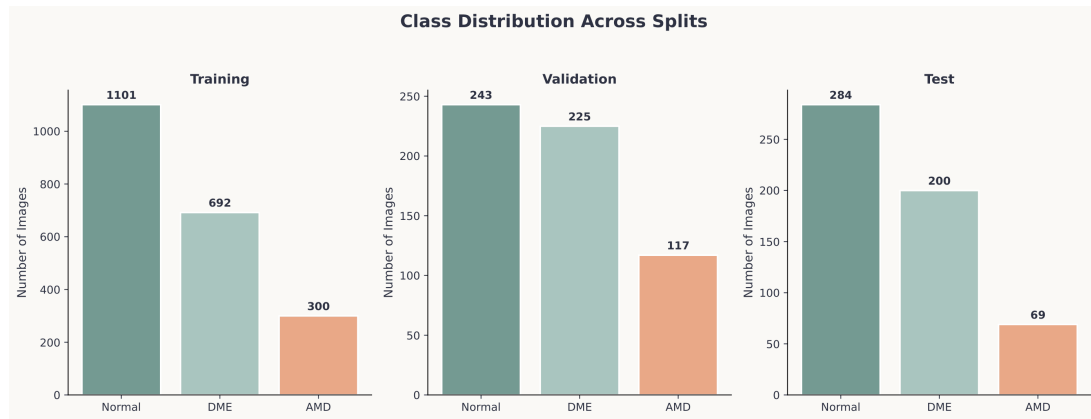


Figure 3.3: Class distribution across the three dataset splits. The relative proportions of AMD, DME, and Normal cases are approximately preserved in each subset, confirming the effectiveness of the stratified splitting strategy.

Chapter 4

Models and Experimental Setup

4.1 Experimental Setup and Reproducibility

All experiments are implemented in Python using PyTorch as the primary deep learning framework. Training and evaluation are performed on a single NVIDIA T4 GPU (16 GB VRAM) via Google Colab. To ensure full reproducibility, a global random seed of 42 is applied at the start of every notebook, fixing the state of Python’s built-in `random` module, NumPy, and all PyTorch random number generators, including the CUDA backend. The `torch.backends.cudnn.deterministic` flag is set to `True` and `benchmark` is disabled, which prevents cuDNN from selecting non-deterministic convolution algorithms. `DataLoader` workers are individually seeded through a `seed_worker` function to guarantee that data loading order is also reproducible across runs.

The same patient-level splitting strategy described in Chapter 3 is applied identically across all models: `StratifiedGroupKFold` with `n_splits = 3` and `n_splits = 2` yields a training set of 77 patients, a validation set of 21 patients, and a test set of 19 patients, with no patient identity shared across partitions.

4.2 CNN Baseline and EfficientNet-Bo

Architecture

Two convolutional models are trained and compared. The first is a lightweight custom CNN built from scratch, serving as a simple baseline. It consists of three convolutional blocks, each containing a convolutional layer, Batch Normalisation, and a ReLU activation, followed by max pooling. A fully connected head with dropout is then applied to produce the final class logits over the three categories (AMD, DME, Normal).

The second model is EfficientNet-B0, initialised with weights pre-trained on ImageNet. The original classification head is replaced with a new linear layer mapping to three output classes. Training follows a two-phase fine-tuning strategy: during the first ten epochs, the convolutional backbone is kept frozen and only the new head is updated; from epoch eleven onwards, all layers are unfrozen and the entire network is trained jointly with a reduced learning rate. This progressive unfreezing reduces the risk of catastrophic forgetting while allowing the backbone to adapt to the OCT domain.

Training Configuration

Both models are trained using the Adam optimiser with an initial learning rate of 1×10^{-3} and weight decay of 10^{-4} . A ReduceLRonPlateau scheduler halves the learning rate when validation loss stops improving for five consecutive epochs. Training is run for a maximum of 50 epochs with early stopping based on validation loss (patience of 10 epochs); the best checkpoint is restored at the end of training. The loss function is cross-entropy weighted by inverse class frequency, computed from the training set using `sklearn.utils.class_weight.compute_class_weight` with the 'balanced' option. This assigns a higher penalty to AMD misclassifications, counteracting the class imbalance described in Section 3.1. Input images follow the preprocessing and augmentation pipeline defined in Chapter 3: grayscale conversion, resizing to 512×512 pixels, and normalisation to $[-1, 1]$, with stochastic augmentations applied exclusively to the training set.

4.3 MedGemma-4B

Adaptation for OCT Classification

MedGemma-4B is a large multimodal model that couples a SigLIP-based vision encoder (MedSigLIP) with a Gemma 3 language model backbone, as described in Section 2.2. End-to-end fine-tuning of a model of this scale is computationally intractable on the available hardware and statistically ill-advised given the size of the OCT5k dataset. The model is therefore used as a frozen feature extractor: the vision encoder is isolated from the rest of the architecture and used to produce a fixed-dimensional embedding for each input image, while a lightweight classifier head is trained on top of these representations.

Model Loading and Memory Management

MedGemma-4B is loaded from the HuggingFace Hub (`google/medgemma-4b-it`) in `float16` precision with automatic device placement. Immediately after loading, the language model and the multimodal projector are explicitly deleted from the checkpoint and GPU memory is released via `torch.cuda.empty_cache()`. This step is deliberate: retaining components that play no role at inference time would occupy several gigabytes of VRAM unnecessarily. Only the `vision_tower` submodule is kept, placed in evaluation mode for the remainder of the pipeline.

The feature dimension is determined at runtime by passing a dummy image through the encoder and inspecting the output: if a `pooler_output` attribute is available, its dimensionality is used; otherwise the mean of the `last_hidden_state` token sequence is adopted as the feature vector. Images are preprocessed exclusively by the official `AutoProcessor` image processor, which handles resizing and normalisation according to the SigLIP input specification. The custom preprocessing pipeline described in Chapter 3 is not applied here; images are loaded as RGB PIL objects and passed directly to the processor.

Feature Extraction

Feature extraction is performed as a single offline pass over the entire dataset before training begins. Due to GPU memory constraints, images are processed one at a time (batch size of 1). For each image, the frozen vision tower produces a feature vector that is immediately moved to CPU RAM. After

processing all splits, the resulting tensors are wrapped in `TensorDataset` objects, fully decoupling the large foundation model from the subsequent training loop. This design means that each training epoch operates entirely on cached feature vectors and requires no interaction with the vision encoder, making iteration fast regardless of model size.

Feature extraction failed for 42 test images due to format incompatibilities during the RGB PIL conversion required by the official `AutoProcessor`. This reduced the effective test set for MedGemma-4B to 511 images; these cases were excluded from the evaluation and are not included in any reported metric for this model.

Classifier Head

The classifier is a two-layer MLP. It takes the extracted feature vector as input, applies a linear projection to a hidden dimension of 512, followed by Layer Normalisation, a GELU activation, dropout with rate 0.3, and a final linear layer mapping to the three output classes. The total number of trainable parameters in the head is small relative to the feature extractor, making this configuration practical even for a dataset of modest size.

Training Procedure

The classifier head is trained for a maximum of 80 epochs using the Adam optimiser with a learning rate of 3×10^{-4} and weight decay of 10^{-4} . A `ReduceLROnPlateau` scheduler halves the learning rate when validation loss does not improve for 5 consecutive epochs. Early stopping with a patience of 10 epochs monitors validation loss and restores the best checkpoint at the end of training. The same inverse-frequency class-weighted cross-entropy loss used for the CNN models is applied here, with weights recomputed from the MedGemma training partition. After training, the classifier weights and a configuration file recording feature dimensions, class mapping, and test metrics are saved to disk, so that inference can be performed without rerunning the full extraction pipeline.

4.4 MIRAGE

Adaptation for OCT Classification

MIRAGE is a Vision Transformer foundation model pre-trained on large-scale paired retinal OCT and SLO data via multimodal masked autoencoding, as described in Section 2.2. Unlike `EfficientNet-B0`, which is fine-tuned end-to-end from the first epoch, and unlike MedGemma, which is used purely as a frozen feature extractor, MIRAGE is adapted through a two-phase training strategy that first trains a linear classification head on a frozen backbone, then unlocks all weights for full fine-tuning. This approach is standard for large ViT-based models: the linear probing phase warms up the classifier head before the backbone is exposed to gradient updates, reducing the risk of destabilising the pre-trained representations.

The specific variant used is `MIRAGEclsGlobal`, instantiated via the official `mirage_wrapper` factory. This variant processes input images as sequences of non-overlapping patches of size 32×32 pixels and produces a global representation by averaging all patch token embeddings, which is then passed to a linear classification head. The pre-trained weights (`MIRAGE-Base.pth`) are downloaded from the official repository and loaded at initialisation.

Preprocessing

MIRAGE requires a different preprocessing pipeline from the one used for the CNN models and MedGemma. Images are loaded as single-channel grayscale, resized to 512×512 pixels using Lanczos resampling, and converted to float32 tensors in the range $[0,1]$ by dividing by 255. A per-image min-max normalisation is then applied:

$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

This step maps each image independently to the range $[0,1]$ and is applied both before and after augmentation on the training set to ensure the input distribution remains consistent after geometric transformations. Validation and test images receive only the initial rescaling and min-max normalisation, without any augmentation.

The augmentation pipeline applied to training images consists of a random horizontal flip with probability 0.5, a random rotation of up to 10° applied with probability 0.5, and a random affine transformation with translation up to 10%, scale in $[0.9, 1.1]$, and shear of 5° , also applied with probability 0.5.

Training Procedure

Training is split into two sequential phases, each running for a maximum of 50 epochs with early stopping based on validation balanced accuracy (patience of 10 epochs, minimum improvement threshold 10^{-3}). Balanced accuracy is used as the early stopping criterion rather than validation loss because it is more robust to class imbalance and directly reflects per-class recall, which is clinically more informative than aggregate cross-entropy. The loss function is the same inverse-frequency class-weighted cross-entropy used for all other models.

Phase 1: Linear Probing. The ViT backbone is fully frozen and only the classification head and the final layer norm are set as trainable. The AdamW optimiser is used with a learning rate of 10^{-3} and weight decay of 10^{-2} . A cosine annealing scheduler reduces the learning rate smoothly to 10^{-8} over the course of the phase.

Phase 2: Full Fine-Tuning. All model parameters are unfrozen and the entire network is trained jointly. The learning rate is reduced to 10^{-5} to avoid large gradient updates that could distort the pre-trained representations. The same AdamW optimiser, cosine annealing scheduler, and early stopping configuration are applied. The best checkpoint from Phase 2 is restored at the end of training.

4.5 Uncertainty Quantification

The uncertainty quantification pipeline implemented in this work applies the methods introduced in Section 5.2, temperature scaling, confidence-based thresholding, and entropy scoring, to all four models. This section details the specific implementation choices and parametrisation adopted for each.

Temperature Scaling

For the CNN and EfficientNet-B0 models, the optimal temperature T is found by grid search over $[0.5, 5.0]$ in 100 uniformly spaced steps, minimising the cross-entropy loss on the full validation set

computed on logits collected in a single forward pass with gradients disabled. For MedGemma, the search range is extended to $[0.5, 10.0]$ over 200 steps to account for the different confidence distribution of the foundation model head. For MIRAGE, the same grid search is applied on logits collected from the validation loader after Phase 2 fine-tuning.

Confidence-Based OOD Detection

Inputs for which the maximum calibrated probability falls below a fixed threshold τ are flagged as uncertain and referred to a human expert. For the CNN and EfficientNet models $\tau = 0.65$; for MedGemma $\tau = 0.70$; for MIRAGE $\tau = 0.65$. All thresholds are applied after temperature scaling.

Entropy Scoring

For a three-class problem, the theoretical maximum entropy is $\log(3) \approx 1.099$, corresponding to a uniform prediction. The entropy distributions of the in-distribution test set and the glaucoma OOD set are compared via histograms, and a threshold is derived as the midpoint between the mean in-distribution validation entropy and the mean OOD (glaucoma) entropy. For MedGemma, an OOD detection AUROC is additionally computed using entropy as the scoring function.

OOD Test Set

The out-of-distribution evaluation uses 100 glaucoma OCT images sampled randomly from SYN-OCT [20], a publicly available synthetic dataset of retinal OCT scans from healthy and glaucoma eyes. The same sample is used consistently across all models to ensure comparability.

Chapter 5

Results and Evaluation

This chapter reports the experimental results obtained from training and evaluating the four models described in Chapter 4 on the OCT5k dataset. All metrics are computed exclusively on the held-out test set, which was never used during training, validation, or hyperparameter selection. Results are organised into three areas: classification performance and calibration analysis, out-of-distribution detection, and visual explanation via saliency maps. For completeness, a comparative discussion across all four models closes the chapter.

5.1 Classification Performance and Calibration

Classification Performance

Table 5.1 summarises the primary classification metrics for all four models on the test set. For all four models, accuracy and macro-averaged AUC are accompanied by 95% bootstrap confidence intervals computed over 1000 resampled draws, following established practice for evaluating medical AI models on small test sets. For MIRAGE-Base, the Matthews Correlation Coefficient (MCC) is additionally reported in the footnote, as it provides a robust measure of classification quality under class imbalance. Macro-averaged AUC is reported for all four models as the primary ranking metric, since it treats each diagnostic class equally regardless of its frequency and is sensitive to discrimination ability across the full range of decision thresholds.

Table 5.1: Summary of classification performance on the held-out test set. Bootstrap 95% confidence intervals (1,000 iterations) are reported for all four models.

Model	Accuracy	95% CI	Macro AUC	95% CI
CustomCNN	0.854	[0.825, 0.882]	0.917	[0.900, 0.934]
EfficientNet-B0	0.957	[0.940, 0.973]	0.995	[0.992, 0.998]
MedGemma-4B	0.900	[0.873, 0.924]	0.979	[0.968, 0.988]
MIRAGE-Base	0.960	[0.944, 0.975]	0.995	[0.992, 0.998]

MIRAGE-Base reports standard accuracy. The Matthews Correlation Coefficient (MCC) is 0.933.

EfficientNet-B0 and MIRAGE-Base achieve the highest overall performance, with macro AUC values of 0.995 and 0.995 respectively, placing both models at near-ceiling discrimination ability across all

three diagnostic categories. Both confidence intervals are narrow and nearly identical: [0.992, 0.998] for EfficientNet-B0 and [0.992, 0.998] for MIRAGE-Base, reflecting highly consistent results across resampling iterations. MIRAGE-Base additionally achieves a standard accuracy of 96.0% [0.944, 0.975], the highest among the four models. MedGemma-4B reaches an accuracy of 90.0% and a macro AUC of 0.979, a strong result given that its visual encoder was kept entirely frozen and only a lightweight two-layer MLP classifier was trained on top of the extracted features. The CustomCNN baseline achieves an accuracy of 85.4% and a macro AUC of 0.917, which is a reasonable outcome for a network of fewer than 100,000 parameters trained from scratch on approximately 2,100 training images, but confirms the advantage that transfer learning and domain-specific pre-training provide over random initialisation in low-data medical imaging settings.

Per-Class Analysis and Confusion Matrices

Tables 5.2 through 5.5 report per-class precision, recall, F1-score, and one-vs-rest AUC for all four models on the test set. The AMD class, which accounts for only 15% of the dataset and 69 test images, deserves particular attention: misclassifying an AMD scan as Normal would mean failing to identify a patient who requires anti-VEGF treatment to prevent irreversible vision loss.

Table 5.2: Per-class metrics for CustomCNN on the test set (n = 553).

Class	Precision	Recall	F1-Score	AUC (OvR)
AMD	0.489	0.333	0.397	0.846
DME	0.806	0.850	0.827	0.924
Normal	0.946	0.982	0.964	0.982
Macro avg	0.747	0.722	0.729	0.917

Table 5.3: Per-class metrics for EfficientNet-B0 on the test set (n = 553).

Class	Precision	Recall	F1-Score	AUC (OvR)
AMD	0.862	0.812	0.836	0.989
DME	0.960	0.955	0.957	0.997
Normal	0.976	0.993	0.984	0.999
Macro avg	0.932	0.920	0.926	0.995

Table 5.4: Per-class metrics for MedGemma-4B on the test set (n = 511).

Class	Precision	Recall	F1-Score	AUC (OvR)
AMD	0.795	0.880	0.835	0.981
DME	0.957	0.810	0.878	0.985
Normal	0.907	0.947	0.926	0.971
Macro avg	0.886	0.879	0.880	0.979

Table 5.5: Per-class metrics for MIRAGE-Base on the test set (n = 553).

Class	Precision	Recall	F1-Score	AUC (OvR)
AMD	0.968	0.870	0.916	0.994
DME	0.933	0.980	0.956	0.995
Normal	0.979	0.968	0.973	0.996
Macro avg	0.960	0.939	0.949	0.995

The AMD class is where the gap between models is most pronounced. CustomCNN achieves an AMD precision of only 0.489 and an AMD recall of 0.333, yielding an F1-score of 0.397. This poor performance on the minority class is not surprising for a model trained from scratch: without pre-trained representations, the network struggles to form robust features for a class with only 300 training images, and the class-weighted loss function partially compensates but cannot fully bridge the gap. In contrast, EfficientNet-B0 reaches an AMD F1 of 0.836, MedGemma-4B achieves 0.835, and MIRAGE-Base achieves 0.916, all representing a substantial improvement that reflects the richer feature representations provided by pre-training or domain-specific foundation model architectures.

An interesting pattern emerges in the confusion matrix for CustomCNN: the model systematically misclassifies AMD scans as DME rather than as Normal. This tendency arises because both AMD and DME involve the presence of fluid in the retinal layers, producing a degree of visual similarity in OCT cross-sections that a shallow network is unable to fully resolve. EfficientNet-B0 and the two foundation models show considerably more consistent discrimination between the two fluid-associated pathologies, suggesting that deeper and richer feature hierarchies are needed to capture the subtle structural differences between subretinal fluid in AMD and intraretinal cystoid spaces in DME.

A related and clinically important observation concerns MedGemma-4B’s AMD recall of 0.880, which is actually the highest across all four models, despite MedGemma having the second-lowest overall accuracy. This reflects a different precision-recall trade-off: the MLP classifier trained on MedGemma features is more sensitive to AMD but somewhat less specific, yielding a lower AMD precision of 0.795 compared to MIRAGE’s 0.968. In a clinical screening scenario, high recall on the minority class is arguably the more critical property, since a false negative on AMD carries greater consequences than a false positive that leads to a follow-up examination.



Figure 5.1: Normalised confusion matrices for all four models on the held-out test set. Each cell reports the fraction of true-class images assigned to each predicted category. Perfect classification corresponds to a diagonal matrix with all entries equal to 1.00. The CustomCNN confusion matrix (top left) shows the largest off-diagonal mass, primarily concentrated in the AMD row, where just over half of AMD images are misclassified as DME.

Training Dynamics

CustomCNN was trained for a maximum of 50 epochs. Early stopping triggered at epoch 49, with the best validation loss of 0.603 recorded at epoch 34. The continued slow improvement in training accuracy after this point while validation loss stagnated indicates mild overfitting in the final training phase, despite the regularisation provided by dropout and weight decay.

EfficientNet-B0 followed a two-phase schedule. During the first ten epochs, the backbone was frozen and only the classification head was updated, serving as a warm-up phase. The backbone was unfrozen at epoch 11 with the learning rate reduced to 10^{-5} , after which the validation loss dropped substantially. The best validation loss of 0.317 was reached at epoch 26 and early stopping triggered at epoch 41. The progressive unfreezing strategy prevented catastrophic forgetting of the ImageNet representations during the initial adaptation phase.

MIRAGE-Base training proceeded in two analogous phases. Linear probing of the frozen ViT backbone ran for a maximum of 50 epochs and stopped at epoch 44, achieving a best validation balanced accuracy of 0.942. Full fine-tuning in Phase 2 ran for a maximum of 50 further epochs and stopped at epoch 25, improving the best validation balanced accuracy to 0.954. Balanced accuracy rather than loss was used as the early stopping criterion in both phases, as it is a more robust measure of per-class performance under class imbalance. MedGemma-4B trained its MLP head for 80 epochs with early

stopping, with the best validation loss reached at epoch 7 before the scheduler began to reduce the learning rate.

ROC Curves

Figure 5.2 shows the one-vs-rest ROC curves for all four models on their respective test sets, with per-class and macro-average AUC values. For both EfficientNet-B0 and MIRAGE-Base, all three class-specific AUC values exceed 0.989 and 0.994 respectively, indicating near-perfect binary discrimination for each diagnostic category. The macro-average AUC values of 0.995 and 0.995 confirm that both models achieve highly reliable probability rankings across the full range of operating thresholds, a property that is directly relevant to the confidence-based out-of-distribution detection mechanism described in Section 5.2.

MedGemma-4B achieves AUC values of 0.981, 0.985, and 0.971 for AMD, DME, and Normal respectively, with the Normal class showing a slightly lower value despite having the largest support in the test set. The overall macro AUC of 0.979 is strong and confirms that the MedSigLIP vision encoder extracts representations that are highly discriminative for retinal OCT classification even without any fine-tuning of the encoder weights.

For CustomCNN, the AMD ROC curve shows the steepest drop in performance relative to the other models: an AUC of 0.846 for AMD compared to 0.924 and 0.982 for DME and Normal. This per-class AUC pattern is directly consistent with the low AMD recall discussed above and confirms that the model’s AMD detector is substantially less reliable than its DME and Normal detectors.



Figure 5.2: One-vs-rest ROC curves on the held-out test set for all four models. Each coloured curve corresponds to one diagnostic class (AMD, DME, Normal); the dashed black line indicates the macro-average ROC. The grey dotted diagonal represents chance-level discrimination. AUC values for each class and the macro average are reported in the legend. EfficientNet-B0 and MIRAGE-Base achieve near-ceiling AUC across all three classes.

Reliability Diagrams

The reliability diagrams generated before and after temperature scaling for all four models are presented in Figure 5.4. In a perfectly calibrated model, the mean predicted confidence within each bin equals the observed accuracy in that bin, placing all points on the diagonal. Deviations below the diagonal correspond to overconfidence (where the model’s confidence exceeds its actual accuracy), while deviations above the diagonal indicate underconfidence.

Before scaling, EfficientNet-B0 and MIRAGE-Base show clear overconfidence in the high-confidence range. For MIRAGE-Base in particular, the uncalibrated predictions are heavily concentrated near 1.0

(as shown by the confidence histograms), yet the actual empirical accuracy is substantially lower. This visually confirms the need for a high temperature ($T = 4.773$). After scaling, the calibration curves for both models move substantially closer to the diagonal.

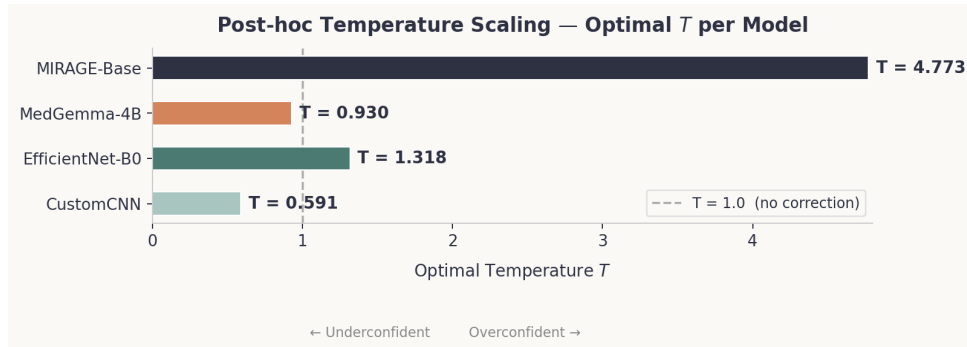


Figure 5.3: Optimal temperature T values calculated for each model. Models with $T > 1$ (MIRAGE-Base, EfficientNet-B0) require post-hoc softening to correct overconfidence, while models with $T < 1$ (CustomCNN, MedGemma-4B) are initially closer to or underconfident relative to the diagonal.

CustomCNN, by contrast, shows a more erratic pattern before scaling, a consequence of the model’s already-noted difficulty in forming robust representations. The effect of temperature scaling on CustomCNN is visible but less drastic, as expected from its lower optimal temperature ($T = 0.591$). Finally, MedGemma-4B ($T = 0.930$) is already well-calibrated before scaling, with its initial curve closely hugging the perfect calibration line and requiring only minor adjustments.



Figure 5.4: Reliability diagrams (calibration curves) for all four evaluated models on the test set, shown before (dashed lines, circles) and after (solid lines, squares) temperature scaling. The diagonal dotted line represents perfect calibration, while histograms at the bottom of each subplot indicate the distribution of predicted confidences. EfficientNet-B0 and MIRAGE-Base exhibit significant overconfidence before scaling, which is effectively corrected post-hoc, whereas CustomCNN and MedGemma-4B are initially closer to the ideal calibration line.

5.2 Out-of-Distribution Detection

Baseline OOD Behaviour Before Calibration

Before temperature scaling is applied, all four models are evaluated on 100 glaucoma OCT images drawn from the SYNOCT dataset. Glaucoma was chosen as the OOD class because it produces structurally distinct OCT patterns, including retinal nerve fibre layer thinning and optic disc cupping, that are not represented in any of the three training categories. The central question is whether the models recognise these images as anomalous or confidently assign them to a known class.

Table 5.6 shows the prediction distributions and mean maximum confidence values before calibration. Two patterns are immediately apparent. First, CustomCNN assigns all 100 glaucoma images to DME with a mean confidence of 0.988, flagging none of them as uncertain under any threshold. This is the most dangerous possible failure mode: the model is maximally confident and completely wrong. The assignment to DME is mechanistically consistent with the confusion matrix pattern noted in Section 5.1, where the model also misroutes AMD to DME. Without rich pre-trained representations, the network appears to have learned DME as a residual catch-all category for any non-Normal scan. MIRAGE-Base shows a markedly different pattern: glaucoma images are distributed across AMD (55%) and Normal (41%), with a mean confidence of 0.801, and 26 images already fall below the 0.65 threshold before any calibration is applied.

Second, EfficientNet-B0 shows substantially more distributional uncertainty even before calibration: glaucoma images are split between DME (57%) and AMD (43%), and the mean maximum confidence drops to 0.701, with 42 images already falling below the 0.65 threshold. MedGemma-4B assigns all glaucoma images to AMD with mean confidence 0.938, flagging 7 images under the 0.70 threshold.

Table 5.6: OOD baseline behaviour on 100 glaucoma images before temperature scaling. Flagging threshold: $\tau = 0.65$ for CNN models and MIRAGE, $\tau = 0.70$ for MedGemma.

Model	Predicted class	Mean conf.	Flagged
CustomCNN	DME (100%)	0.988	0/100
EfficientNet-B0	DME (57%) / AMD (43%)	0.701	42/100
MedGemma-4B	AMD (100%)	0.938	7/100
MIRAGE-Base	AMD (55%) / Normal (41%)	0.801	26/100

Effect of Temperature Scaling and Entropy-Based Detection

After temperature scaling, a complementary entropy-based detection signal is computed for each model. The Shannon entropy of the predicted probability distribution over three classes ranges from 0 for a perfectly peaked prediction to $\log(3) \approx 1.099$ nats for a uniform one. A per-model entropy threshold is derived as the midpoint between the mean in-distribution validation entropy and the mean OOD (glaucoma) entropy. Inputs whose entropy exceeds this threshold are flagged as potentially out-of-distribution.

Table 5.7 summarises the flagging results after calibration, together with the OOD detection AUROC computed by treating validation images as in-distribution (label 0) and glaucoma images as out-of-distribution (label 1), and using calibrated entropy as the scoring function.

Table 5.7: OOD detection performance after temperature scaling. Val entropy mean is the mean Shannon entropy of the in-distribution validation set at temperature T . Glaucoma entropy mean is the corresponding value for the 100 OOD images. AUROC measures the separability of the two distributions using entropy as a scoring function.

Model	T	Val entropy	OOD entropy	Flagged	AUROC
CustomCNN	0.591	0.543	0.005	0/100	0.030
EfficientNet-B0	1.318	0.296	0.713	87/100	0.872
MedGemma-4B	0.930	0.425	0.177	7/100	0.204
MIRAGE-Base	4.773	0.278	0.969	96/100	0.977

The results reveal a sharp divergence in OOD detection ability across models. Among the four models, MIRAGE-Base and EfficientNet-B0 emerge as the most effective OOD detectors. For EfficientNet-B0, after scaling, the mean glaucoma entropy rises to 0.713 nats, well above the in-distribution mean of 0.296 and the derived threshold of 0.505, resulting in 87 out of 100 glaucoma images being flagged. The OOD AUROC of 0.872 confirms that the entropy distributions of in-distribution and OOD inputs are well separated. Figure 5.5 shows this separation visually: the glaucoma entropy histogram is shifted substantially to the right of the in-distribution histogram for EfficientNet, while for CustomCNN the two distributions are almost entirely reversed, with glaucoma producing lower entropy than the test set due to its extreme overconfidence.

MIRAGE-Base achieves the strongest OOD detection performance of all four models, with an AUROC of 0.977 and 96 out of 100 glaucoma images flagged after temperature scaling ($T = 4.773$). The mean OOD entropy reaches 0.969 nats, significantly higher than the in-distribution validation mean of 0.278 nats. Even before calibration, MIRAGE already showed distributional uncertainty on glaucoma inputs, splitting predictions across AMD and Normal with a mean confidence of only 0.801. Temperature scaling amplifies this uncertainty substantially, reducing mean confidence to 0.513 and enabling near-complete flagging of the OOD set.

The two weakest detectors are CustomCNN and MedGemma-4B. CustomCNN’s failure is total: an AUROC of 0.030 means that the entropy scoring function performs nearly at chance in reverse, with glaucoma images consistently producing lower entropy than in-distribution images. This is a direct consequence of the model collapsing all glaucoma inputs to a single class with near-certainty, leaving no distributional signal for the detector to exploit. MedGemma-4B’s failure is subtler. Despite strong classification performance, the MLP classifier trained on frozen MedSigLIP features assigns glaucoma images to AMD with high confidence and low entropy (mean 0.177 nats), below the in-distribution entropy mean of 0.425. The entropy score therefore fails to separate the two distributions, yielding an AUROC of 0.204. This result illustrates a fundamental limitation of entropy-based detection: if a foundation model has learned representations in which unseen pathologies project near an existing class centroid, the resulting predictions will be confident and low-entropy regardless of calibration.

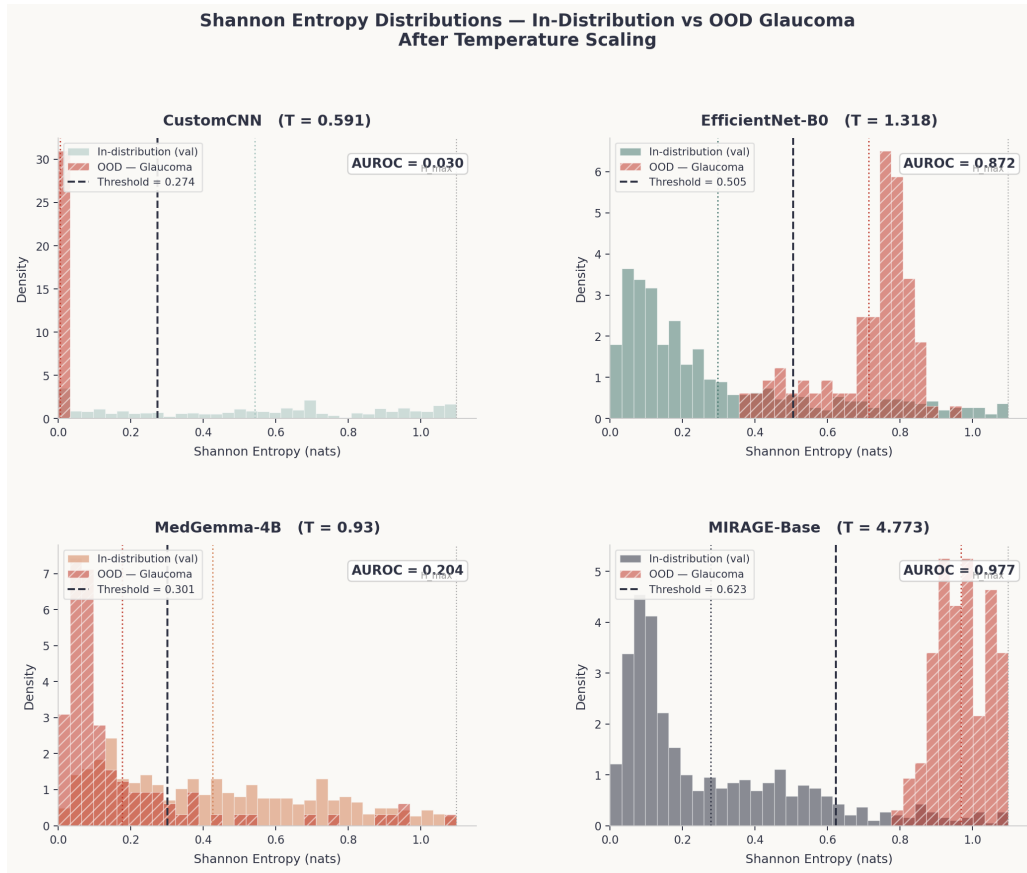


Figure 5.5: Shannon entropy distributions for in-distribution test images (blue) and glaucoma OOD images (red) after temperature scaling, for all four models. The vertical dashed line marks the entropy flagging threshold for each model. For EfficientNet-B0 the two distributions are clearly separated, with glaucoma images concentrated in the high-entropy region. For CustomCNN the distributions are reversed: glaucoma images produce lower entropy than in-distribution images, explaining the near-zero AUROC.

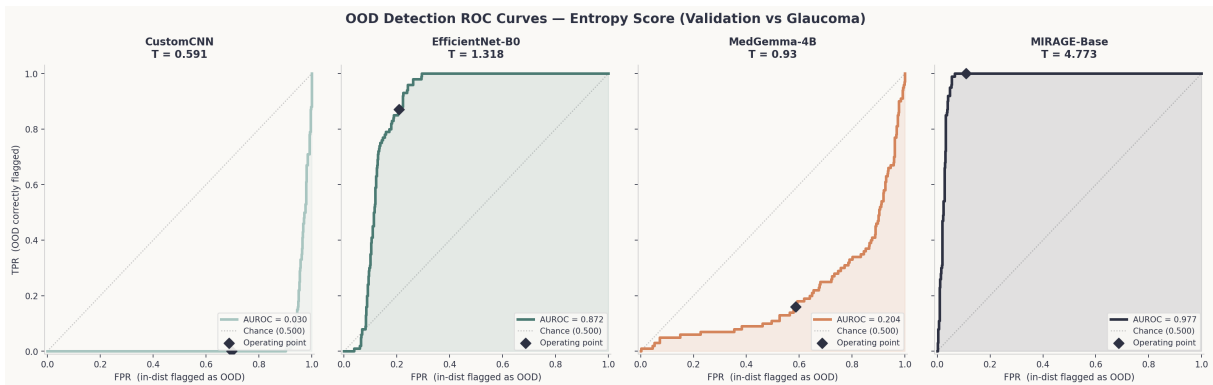


Figure 5.6: OOD detection ROC curves for all four models, using calibrated Shannon entropy as the scoring function. The area under each curve (AUROC) measures how well the entropy score separates in-distribution validation images from glaucoma OOD images. EfficientNet-B0 (AUROC = 0.872) and MIRAGE-Base (AUROC = 0.977) achieve strong detection performance, while CustomCNN (AUROC = 0.030) and MedGemma-4B (AUROC = 0.204) fail to separate the two distributions reliably.

5.3 Visual Explanations

Grad-CAM for Convolutional Models

Gradient-weighted Class Activation Mapping is applied to both CustomCNN and EfficientNet-B0 to produce spatial explanations for their predictions on the test set. For CustomCNN, the target layer is the third convolutional block (`model.features[-4]`), and for EfficientNet-B0 it is the final convolutional stage of the backbone (`model.base.features[-1]`). In both cases, the Grad-CAM heatmap is computed with respect to the predicted class, so that the highlighted regions correspond to the spatial locations that most positively influenced the model’s classification decision.

Figure 5.7 shows Grad-CAM overlays for five randomly selected test images, displayed for both models. Several observations are consistent across samples. For Normal retinas, both models concentrate attention on the foveal pit and the inner retinal layers immediately surrounding it, which represent the anatomically cleanest signal of a healthy retinal architecture. For DME images, attention is distributed across the regions of intraretinal fluid accumulation, visible in the original OCT as hyporeflective cystoid spaces within the retinal layers. For AMD images, EfficientNet-B0 shows more focused attention on the subretinal space and the region of the retinal pigment epithelium where drusen deposits and choroidal neovascularisation are located, whereas CustomCNN produces broader and less anatomically specific heatmaps. This difference is consistent with the per-class performance gap between the two models: EfficientNet has learned more discriminative AMD-specific features, reflected in both its higher AMD F1-score and its more spatially targeted saliency maps.

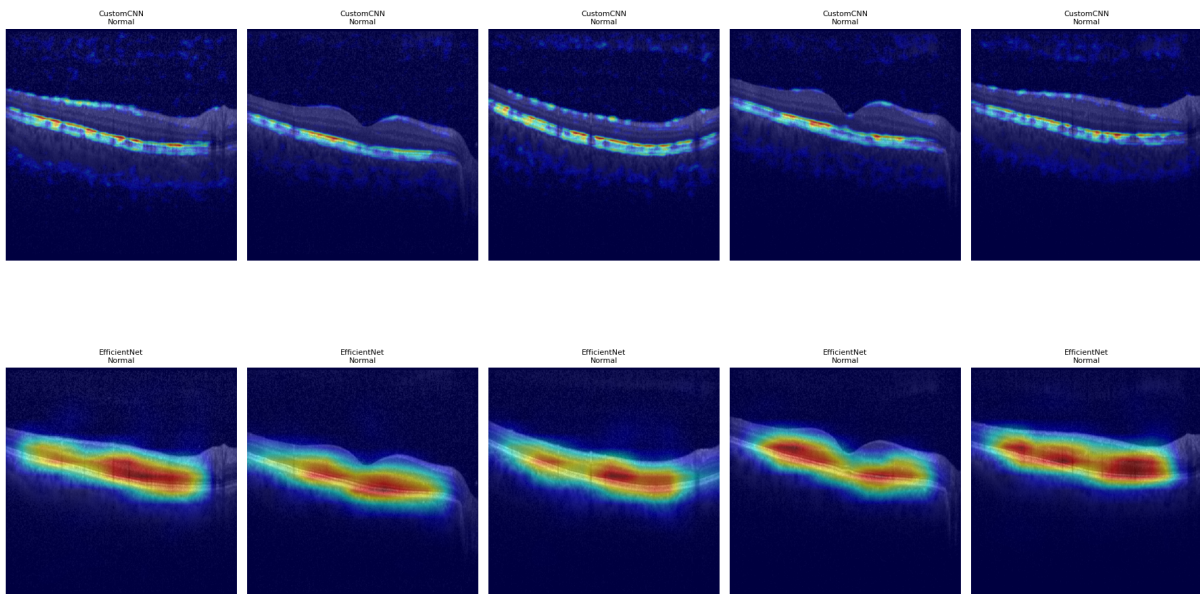


Figure 5.7: Grad-CAM saliency maps for five randomly selected test images. Top row: CustomCNN. Bottom row: EfficientNet-B0. Each image shows the heatmap overlaid on the original OCT scan, with warmer colours indicating regions that most strongly influenced the predicted class. The true class label is reported in the subtitle of each column. EfficientNet-B0 produces more anatomically focused attention than CustomCNN, particularly for AMD images where the relevant pathological features are spatially concentrated in the subretinal region.

Figure 5.8 shows Grad-CAM maps for five glaucoma images from the SYNOCT dataset, again for both models. Because glaucoma images are out-of-distribution, these maps reveal where the models look when forced to assign an unknown pathology to a known class. CustomCNN, which assigns

all glaucoma images to DME with near-certainty, produces heatmaps that concentrate on the central retinal layers in a pattern qualitatively similar to its DME attention maps on in-distribution data. EfficientNet-B0, which shows substantially more uncertainty on glaucoma images, produces more diffuse and variable heatmaps, with attention distributed across non-specific regions of the scan rather than concentrated on any anatomically coherent structure.

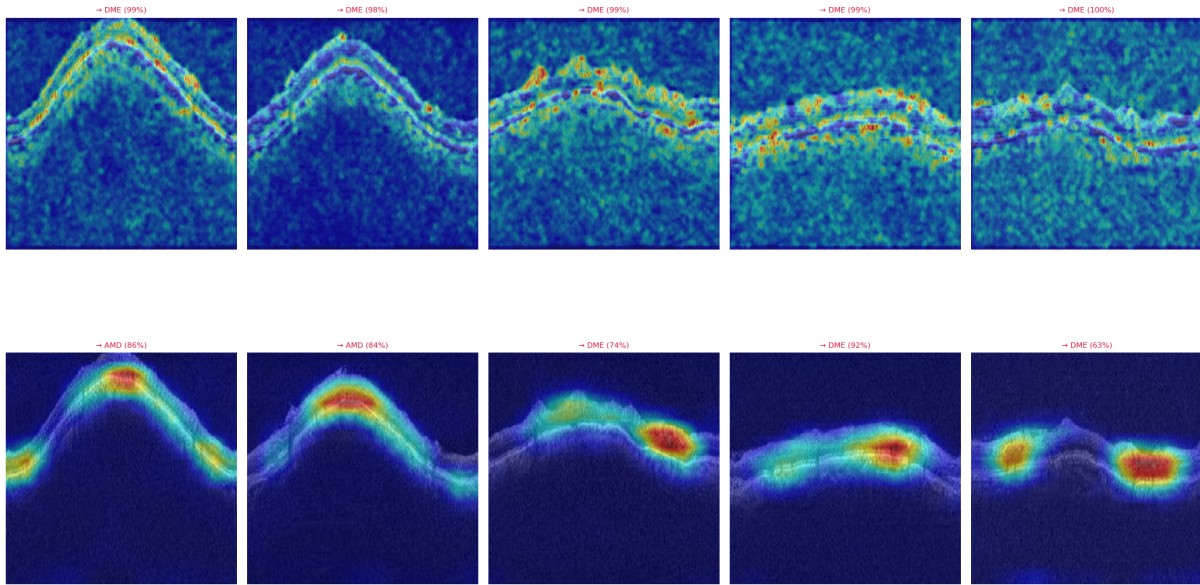


Figure 5.8: Grad-CAM saliency maps for five glaucoma OOD images from the SYNOCT dataset. Top row: CustomCNN (all predicted as DME with high confidence). Bottom row: EfficientNet-B0 (predictions split between DME and AMD).

Occlusion Sensitivity for MIRAGE-Base

Grad-CAM is not applicable to MIRAGE-Base because the model is built on a Vision Transformer backbone. In a ViT, the image is partitioned into non-overlapping 32×32 pixel patches that are processed as a sequence of flat token embeddings. Unlike convolutional feature maps, which retain explicit spatial dimensions, the intermediate representations in a transformer do not correspond to localisable image regions in a way that the Grad-CAM gradient computation can exploit reliably. Applying Grad-CAM to a ViT typically produces heatmaps dominated by global token statistics rather than spatially discriminative regions, yielding maps that are difficult to interpret clinically.

Occlusion sensitivity is used instead. The method systematically replaces each 32×32 patch in the input image with a zero-valued mask and measures the resulting drop in the model’s predicted probability for the target class. A large drop indicates that the occluded region was important for the prediction; a small drop indicates that the region was not used. The resulting importance values are normalised to $[0, 1]$ and resized to the full image resolution using bilinear interpolation, producing a saliency map that is comparable in interpretation to a Grad-CAM overlay.

Figure 5.9 shows occlusion sensitivity maps for one representative image from each of the three diagnostic classes. For the AMD sample, the highest sensitivity is concentrated in the subretinal and sub-RPE region, consistent with the location of drusen and CNV that characterise advanced AMD in OCT. For DME, sensitivity peaks are distributed across the intraretinal layers where fluid pockets disrupt the normal laminar architecture. For the Normal retina, the most sensitive regions correspond

to the central foveal depression and the inner plexiform layer, which constitute the primary visual reference for a healthy scan in the absence of pathological features. In all three cases, the occlusion sensitivity map is spatially coherent and anatomically plausible, providing evidence that MIRAGE-Base bases its predictions on clinically relevant image regions rather than spurious background features.

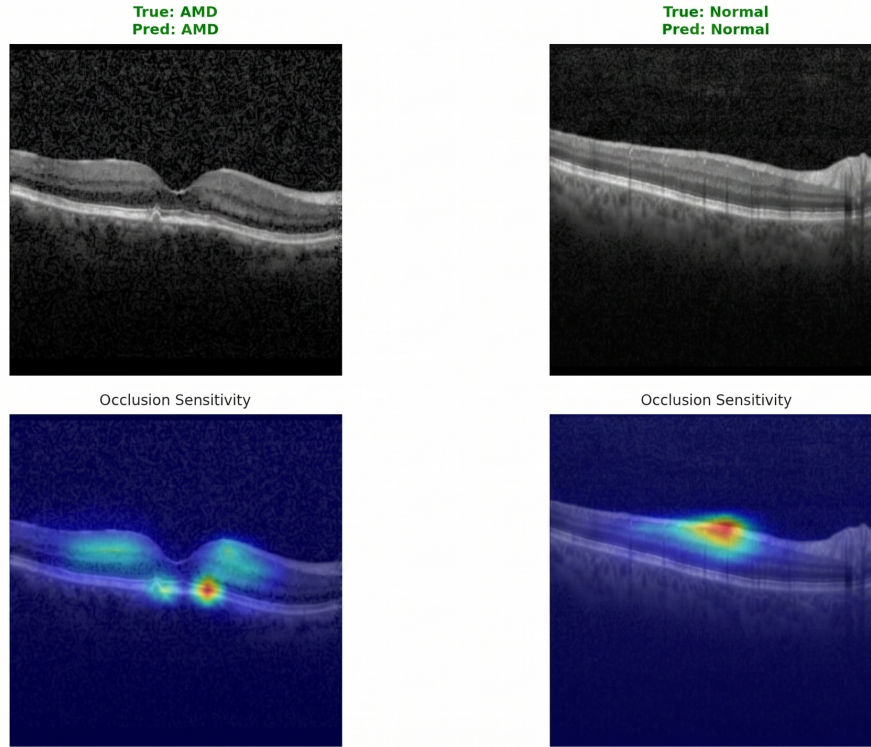


Figure 5.9: Occlusion sensitivity maps for MIRAGE-Base, computed using 32×32 patch masking. Top row: original OCT images for one representative sample from each class (AMD, DME, Normal). Bottom row: corresponding sensitivity overlays, where warmer colours indicate patches whose occlusion causes the largest drop in predicted class confidence. The title of each column indicates the true and predicted class; correctly classified samples are shown in green. Sensitivity is anatomically focused in all three cases.

Explainability Limitations

MedGemma-4B does not support spatial explanation methods in this experimental setup. Because the visual encoder is used as a frozen feature extractor and only the two-layer MLP classifier is trained, the classification decision is made on a single fixed-dimensional embedding vector rather than on spatially organised feature maps or patch tokens. This architectural choice makes it impossible to backpropagate the classifier’s gradient to specific image regions, as the upstream encoder is treated as a black box during both training and inference. Producing meaningful spatial explanations for MedGemma would require either fine-tuning the encoder end-to-end, or applying model-agnostic attribution methods such as LIME or SHAP at the image level, both of which are beyond the scope of this thesis.

It is also important to note that Grad-CAM and occlusion sensitivity, while useful tools for qualitative verification, are not validated explanations in a clinical sense. Both methods highlight regions that are correlated with the predicted class score, but they do not guarantee that the highlighted regions are the same ones that a trained ophthalmologist would identify as diagnostically relevant. A rigorous

evaluation of explanation quality would require expert clinician assessment, which is acknowledged as a necessary direction for future work.

5.4 Comparative Discussion

This section synthesises the results across all four evaluation dimensions to address the central comparative question of this thesis: how do standard deep learning models trained from scratch or adapted via transfer learning compare to medical vision foundation models for retinal OCT classification, and what are the implications for each dimension of trustworthy clinical AI?

Classification Performance

On the primary classification task, EfficientNet-B0 and MIRAGE-Base are effectively equivalent at the level of macro AUC, with values of 0.995 and 0.995 respectively and overlapping performance across all three diagnostic classes. The practical difference between the two models lies in their AMD recall: MIRAGE achieves 0.870 compared to EfficientNet’s 0.812. In a clinical screening context, where the cost of a missed AMD diagnosis substantially exceeds the cost of a false positive, this gap is clinically meaningful and constitutes a genuine advantage of the domain-specific foundation model over transfer learning from a general-purpose source.

MedGemma-4B occupies an interesting position in the comparison. Despite using a frozen encoder and training only a lightweight MLP classifier, it achieves a macro AUC of 0.979 and the highest AMD recall of all four models (0.880). This result demonstrates that a medical vision foundation model can produce embeddings of sufficient quality for near-competitive classification without any adaptation of the visual representations themselves. The cost of this approach is a lower AMD precision (0.795), reflecting that the MLP classifier makes more false positive AMD predictions than the fine-tuned models.

CustomCNN confirms the well-established finding that training from scratch on small medical imaging datasets produces substantially weaker minority-class performance than any form of pre-training. Its AMD F1-score of 0.397 is less than half of the next-weakest model, and its systematic confusion of AMD with DME reveals the inability of a shallow network to learn the fine structural distinction between subretinal and intraretinal fluid without rich pre-trained representations.

Calibration

The calibration results produce a counterintuitive ranking. MIRAGE-Base, despite being the strongest classifier overall, requires the largest temperature correction ($T = 4.773$) of any model. This reflects a general property of large pre-trained transformers: the breadth and depth of the pretraining objective encourages the model to produce strongly peaked predictions, which translate into systematic overconfidence when the model is adapted to a narrow target domain. EfficientNet-B0 also requires softening ($T = 1.318$), consistent with the known overconfidence of fine-tuned convolutional networks. CustomCNN and MedGemma-4B are the best-calibrated models before any correction, with temperatures near or below 1. In the case of CustomCNN, however, near-calibration in the aggregate sense is misleading: the model is uncertain about AMD specifically, and its overall calibration hides class-level miscalibration that is only visible in the per-class reliability diagrams.

Out-of-Distribution Detection

The OOD detection results reveal the most striking inversion of expectations in this comparison. MIRAGE-Base achieves the strongest OOD detection of all four models (AUROC 0.977, 96 images flagged), a result driven both by its naturally more distributed glaucoma predictions before calibration and by the substantial softening effect of its large temperature correction ($T = 4.773$). EfficientNet-B0 follows closely (AUROC 0.872, 87 images flagged). MedGemma-4B and CustomCNN both fail as OOD detectors, for different but structurally similar reasons: both models collapse all glaucoma inputs to a single class with high confidence, leaving no entropic signal for the detector to use.

The key factor that distinguishes effective from ineffective OOD detectors in this setting is not overall classification performance, but the shape of the confidence distribution on uncertain inputs. EfficientNet-B0 distributes its glaucoma predictions across two classes (57% DME, 43% AMD), which produces genuinely uncertain probability distributions with high entropy. CustomCNN and MedGemma-4B, despite having no information about the correct answer, assign all glaucoma images to a single class, producing peaked distributions that the entropy score cannot distinguish from in-distribution predictions.

Summary

No single model dominates across all four evaluation dimensions. EfficientNet-B0 combines strong classification performance with the most reliable OOD detection and clear spatial explainability via Grad-CAM, making it the most balanced choice for a clinically deployable system. MIRAGE-Base matches it on classification AUC and surpasses it on OOD detection, but requires substantial post-hoc calibration and offers only occlusion-based explanation. MedGemma-4B demonstrates that a frozen medical foundation model can achieve competitive classification without any visual encoder adaptation, at the cost of limited explainability and poor OOD sensitivity. CustomCNN establishes the lower bound across all dimensions and confirms that training from scratch on a small medical imaging dataset is insufficient for reliable minority-class performance or uncertainty awareness.

Chapter 6

Prototype Development, Regulatory Framework, and Ethics

6.1 The ORCA Prototype

The experimental pipeline described in the preceding chapters was consolidated into a functional web-based prototype called ORCA (OCT Retinal Classification Assistant), implemented using the Streamlit framework. The prototype is not a clinical product, and it does not claim to be one. It is a research demonstrator that integrates the classification models, uncertainty quantification mechanisms, and visual explanation tools developed in this thesis into a single interactive interface, with the explicit goal of illustrating how these technical components would translate into a decision-support workflow visible to a clinical user.

System Architecture and Interface

The application is structured around two main components. The primary classification page allows a user to upload a retinal OCT scan in PNG, JPEG, or TIFF format, select one of the four models, and configure the confidence threshold and uncertainty estimation mode before running inference. The second component is an informational page providing clinical background on AMD and DME, a description of the classification pipeline, and a summary of the regulatory context governing AI-based diagnostic tools of this type, including the EU MDR and AI Act classifications discussed in Section 6.2.

The inference pipeline is implemented in two modules. The loader module handles model architecture definitions and weight loading, with fallback to a demo mode when trained weights are unavailable. The predictor module implements preprocessing, single-pass inference, Monte Carlo Dropout uncertainty estimation, out-of-distribution detection, and saliency map generation. All preprocessing applied at inference time is identical to the pipeline used during training: grayscale conversion, resizing to 512×512 pixels using Lanczos resampling, and normalisation to the range $[-1, 1]$.

Uncertainty Quantification in the Interface

A distinctive feature of the ORCA interface is its support for Monte Carlo Dropout inference as the default uncertainty estimation mode. When enabled, the system runs 20 stochastic forward passes with dropout active, computes the mean and standard deviation of the resulting probability distributions,

and derives three independent uncertainty signals: the maximum softmax confidence, the Shannon entropy of the mean prediction, and the mean per-class standard deviation across passes. Out-of-distribution detection is triggered when the maximum softmax confidence falls below its threshold ($\tau = 0.65$) or when either the entropy or the mean standard deviation exceeds its respective limit (0.70 nats and 0.12, respectively). Predictions that trigger one or more signals are flagged with a visual alert indicating that the image may show a pathology outside the training distribution and that human review is recommended. Predictions that pass all checks are displayed with a confirmation that the uncertainty signals are within acceptable bounds.

Visual Explanation and Transparency

For each prediction, the interface offers an optional spatial explanation, generated on demand and displayed as a heatmap superimposed on the original OCT scan, provided the selected model supports it. Specifically, Grad-CAM is used for the convolutional models (CustomCNN and EfficientNet-B0), while occlusion sensitivity is applied for the transformer-based MIRAGE-Base; MedGemma-4B does not currently support spatial explainability due to its frozen encoder architecture. Where available, the colour mapping follows the standard convention, with warmer colours indicating regions that most strongly influenced the predicted class score. Users are explicitly instructed to inspect whether the highlighted regions are anatomically consistent with the predicted diagnosis, treating the generated visualisations not as an autonomous explanation but as a prompt for critical engagement with the model's output.

The interface also exposes a preprocessing visualisation that shows the four stages applied to each uploaded image, from the original input through grayscale conversion, resizing, and normalisation, making the transformation pipeline transparent to non-specialist users. A technical details panel at the bottom of the page reports the device in use, the number of model parameters, the full set of OOD thresholds, and a model card summarising the training dataset, supported diagnostic classes, known out-of-scope inputs, and intended use. The persistent footer disclaimer, visible on every page, states explicitly that the tool is intended for research purposes only and does not replace the judgement of a qualified ophthalmologist.

6.2 Regulatory Classification: ORCA as a SaMD

Software systems generating diagnostic classifications from medical imaging fall under the legal category of *Software as a Medical Device* (SaMD) [21]. Because ORCA is intended to support diagnosis of pathologies carrying a risk of irreversible vision loss (AMD and DME), its translation into a clinical product would trigger stringent regulatory pathways.

The Dual European Framework

Within the European Union, ORCA would be governed by two converging legal instruments. Under the *Medical Device Regulation* (EU MDR 2017/745), software providing diagnostic support for high-risk retinal conditions is classified as a **Class IIa** or **Class IIb** device (Rule 11), requiring Notified Body conformity assessments and post-market surveillance. Concurrently, the 2024 EU *Artificial Intelligence Act* horizontally designates all medical AI as *high-risk*, imposing strict mandates on training data governance, technical robustness, and mandatory human oversight [22].

A parallel risk-based approach governs the US market under the FDA framework, where novel diagnostic SaMDs generally require either a 510(k) equivalence filing or a De Novo request, supported by a *Predetermined Change Control Plan* to manage post-deployment model updates [23].

Compliance Gaps at the Prototype Stage

Developed exclusively on the publicly available OCT5k dataset using a single scanner architecture (Heidelberg SD-OCT), ORCA currently operates strictly as a Level-0 research demonstrator. It lacks the formal artifacts mandated for clinical qualification: an ISO 14971 risk management file, a prospective multi-centre Clinical Evaluation Report (CER), and a Technical Documentation File. Explicitly acknowledging these boundaries is a prerequisite for scientific integrity.

6.3 Trustworthiness Evaluation via the FUTURE-AI Framework

To systematically audit the prototype against emerging international standards, ORCA was mapped onto the consensus guidelines of the FUTURE-AI framework [24]. Table 6.1 synthesizes this alignment across its six core pillars.

Table 6.1: System alignment matrix mapping the ORCA prototype against the six core pillars of the FUTURE-AI framework [24].

FUTURE-AI Pillar	ORCA Implementation & Current Limitations
Fairness	Limited. OCT5k lacks demographic metadata; subgroup error disparity cannot be audited.
Universality	Limited. Validated strictly on Heidelberg SD-OCT; zero-shot hardware shifts untested.
Traceability	Partial. Runtime uncertainty quantification implemented; continuous production telemetry missing.
Usability	Partial. Clinician-in-the-loop workflow via spatial explanations; real-world ergonomic trials required.
Robustness	Partial. Photometric augmentations and OOD module active; rare retinal pathologies unrepresented.
Explainability	Implemented. On-demand saliency maps provided as exploratory verification prompts.

Fairness and Universality

Fairness requires identifying and mitigating systematic performance disparities across demographic subgroups [24]. Here, ORCA faces an intractable upstream limitation: the OCT5k dataset contains zero patient metadata (age, sex, ethnicity, or systemic comorbidities). Consequently, potential algorithmic bias remains unquantifiable at this stage [25]. **Universality** demands generalizability across varied clinical environments. ORCA’s training was restricted to Heidelberg Spectralis scans; its zero-shot performance on alternative hardware widely deployed in clinical practice (e.g., Zeiss Cirrus or Topcon Maestro) remains untested, leaving the pipeline vulnerable to well-documented optical coherence speckle and resolution shifts [3].

Traceability and Usability

Traceability involves lifecycle documentation and runtime quality control. ORCA satisfies this partially via runtime uncertainty quantification: rather than forcing a definitive classification, the interface flags high-entropy or low-confidence samples for specialist diversion. However, it lacks continuous production logging. **Usability** dictates that AI must smoothly integrate into clinical ergonomics. By coupling predictions with spatial heatmaps, ORCA intentionally maintains the clinician as the cognitive authority. Yet, true usability requires empirical validation; whether spatial explanations genuinely calibrate clinician trust or inadvertently foster confirmation bias remains an open empirical question [26].

Robustness and Explainability

Robustness requires resilience against real-world data perturbations. While ORCA’s data augmentation mitigates photometric noise, and its OOD module successfully intercepts synthetic glaucoma scans, real-world robustness against unrepresented retinal pathologies (e.g., macular holes or epiretinal membranes) cannot be guaranteed. **Explainability** mandates clinically meaningful transparency. ORCA relies on spatial explainability methods; however, mathematical critiques demonstrate that saliency maps can highlight anatomically plausible structures even when the underlying neural weights rely on spurious correlations [26]. Explainability in ORCA must therefore be treated as an exploratory prompt rather than a diagnostic proof [27].

6.4 Ethical Tensions in Clinical Deployment

Clinical Autonomy and Automation Bias

Although ORCA is strictly structured as an advisory tool, labeling software as “decision support” does not immunize clinical workflows against *automation bias*, the heuristic tendency of humans to over-rely on automated prompts during high-throughput, fatigue-inducing tasks [28]. In high-volume ophthalmology clinics, a confident false-positive generated by ORCA could anchor a clinician’s judgment, transforming a nominal “human-in-the-loop” into a passive rubber stamp.

Data Privacy under the GDPR

While model training utilized anonymized open-access data, real-world inference transforms uploaded OCT scans into highly sensitive biometric data under Article 9 of the European GDPR. Because deep learning representations can inadvertently encode latent biomarkers (e.g., biological age or cardiovascular risk factors not intended for retinal diagnosis) [29], deployment requires strict ephemeral processing pipelines where images are purged from GPU memory immediately post-inference.

Epistemic Limits and Residual Risk

Out-of-distribution thresholding provides a useful safety net, but it does not guarantee absolute certainty. As established by Hendrycks and Gimpel [15], softmax-derived signals can exhibit pathological overconfidence when confronting domain shifts radically distant from the manifold of the validation set. Ethically, deploying ORCA requires establishing unambiguous governance frameworks defining institutional liability when an unflagged OOD sample results in diagnostic injury.

6.5 Pathway toward Clinical Deployment

Transforming the ORCA prototype into a certified clinical software requires the following four steps:

1. **Hardware-Heterogeneous Validation:** Prospective evaluation across multi-centre cohorts utilizing non-Heidelberg interferometers to quantify and patch cross-scanner domain shift.
2. **Metadata-Linked Auditing:** Re-benchmarking the architecture on demographically annotated datasets to satisfy FUTURE-AI fairness mandates.
3. **In-Silico and In-Vivo Ergonomic Trials:** Measuring diagnostic accuracy, time-to-decision, and susceptibility to automation bias in a cohort of practising ophthalmologists using ORCA.
4. **Dynamic Drift Governance:** Implementing automated telemetry to track production entropy distributions, establishing clear statistical triggers for model retraining.

Chapter 7

Discussion and Conclusion

7.1 Summary of Contributions

This thesis set out to answer a practical question: can an AI-based decision support system for retinal OCT classification be built that is not only accurate, but also calibrated, safety-aware, and interpretable enough to be taken seriously as a clinical tool? The answer, at least at the prototype level, is yes, but with important caveats that depend heavily on which model is chosen and which property of trustworthiness is prioritised.

ORCA was developed as a concrete instantiation of this question. Four models were trained, calibrated, and evaluated on the OCT5k dataset under a strict patient-level splitting strategy that prevents data leakage, a methodological choice that is often overlooked in the medical imaging literature but is essential for credible generalisation estimates. Classification performance, confidence calibration, out-of-distribution detection, and spatial explainability were evaluated independently, rather than treating accuracy as a sufficient proxy for all four dimensions.

7.2 Deep Learning vs Foundation Models: Comparative Analysis

The central comparative question of this thesis was whether medical vision foundation models offer a meaningful advantage over standard transfer learning for retinal OCT classification. The results show that there is no simple answer; it depends entirely on which aspect of the model we are evaluating.

On classification performance, EfficientNet-B0 and MIRAGE-Base are effectively indistinguishable at the level of macro AUC (0.995 each), which is a result that deserves reflection. MIRAGE was pre-trained on over 260,000 paired retinal OCT and SLO images from a specialised clinical dataset, while EfficientNet-B0 was pre-trained on ImageNet, a collection of everyday photographs. The fact that the gap between them collapses to a difference in AMD recall (0.870 vs 0.812) rather than a wholesale performance advantage for the domain-specific model suggests that, for a three-class closed-world classification task on a single scanner type, the added value of domain-specific pre-training is concentrated in the minority class rather than spread uniformly across categories. MedGemma-4B, used purely as a frozen feature extractor with no adaptation of the visual encoder, achieves a macro AUC of 0.979 and the highest AMD recall of all four models (0.880), which is perhaps the most surprising single result in this work. It demonstrates that the representations learned by a large medical vision

foundation model are already discriminative enough for near-competitive performance without any gradient flowing into the encoder at all.

The calibration dimension produces the most counterintuitive result. MIRAGE-Base, the strongest classifier, requires a temperature correction of $T = 4.773$, the largest of any model in this comparison. This is not a failure of MIRAGE but a property of large pre-trained transformers in general: deep domain-specific pre-training appears to make the model extremely confident in its outputs, which translates into systematic overconfidence when it is adapted to a narrow downstream task. The practical implication is that foundation models and transfer learning models cannot be trusted to be well-calibrated out of the box, and post-hoc calibration is not optional if confidence estimates are going to be used for clinical decision-making or OOD flagging.

The OOD detection dimension reveals the sharpest and most instructive inversion. The two best OOD detectors are MIRAGE-Base (AUROC 0.977) and EfficientNet-B0 (AUROC 0.872), while the two worst are MedGemma-4B (AUROC 0.204) and CustomCNN (AUROC 0.030). What separates effective from ineffective detectors is not classification accuracy or model size, but whether the model distributes its predictions across multiple classes when presented with an unknown input, or collapses them onto a single class with high confidence. CustomCNN treats every non-Normal scan as DME regardless of what it sees; MedGemma-4B treats every glaucoma image as AMD because the learned embedding places glaucoma representations near the AMD centroid. Both behaviours leave no entropic signal for the detector to exploit. This finding has a direct implication for clinical deployment: choosing a model for a safety-critical application requires evaluating its OOD behaviour explicitly, not inferring it from classification performance.

7.3 Limitations

Several limitations of this work deserve explicit acknowledgement. The evaluation is restricted to a single dataset acquired with a single scanner model, which means that the results cannot be assumed to hold across other OCT hardware or patient populations. The OCT5k dataset contains no demographic metadata, making it impossible to audit the models for subgroup fairness or to identify which patient groups might be disproportionately affected by classification errors. The OOD evaluation covers only one unseen pathology (glaucoma from SYNOCT), and it is likely that other retinal conditions such as macular hole or epiretinal membrane would produce very different OOD detection profiles across the four models.

7.4 Future Directions

The most immediate next step is external validation on multi-centre, multi-device data, which is a prerequisite for any regulatory pathway and would allow a meaningful test of whether the performance observed on OCT5k holds under real-world distribution shift. A usability study with practising ophthalmologists would establish whether the spatial explanations and uncertainty signals provided by ORCA are actually useful in a clinical workflow, or whether they introduce new sources of confusion or overreliance.

Broadening the OOD evaluation to include a wider range of unseen pathologies would make the safety layer more meaningful. A single OOD test set is not sufficient to characterise the full range

of failure modes a deployed system might encounter, and future work should treat OOD robustness as a systematic benchmark rather than a single experiment. On the modelling side, the finding that MedGemma’s frozen encoder already achieves strong classification performance suggests that the value of foundation models for this task may lie less in their classification accuracy and more in properties such as cross-modality generalisability and data efficiency that were not directly tested here.

7.5 Closing Remarks

This thesis demonstrates that translating a deep learning model into a reliable clinical decision support system requires evaluating accuracy, calibration, out-of-distribution safety, and explainability as distinct and independent dimensions. Raw classification performance, while necessary, does not inherently guarantee that a model is well-calibrated or safe for real-world deployment.

Furthermore, the findings indicate that larger, domain-specific foundation models do not automatically outperform standard architectures across all criteria. Notably, EfficientNet-B0, demonstrated the most balanced overall profile among the evaluated approaches, combining high discrimination ability with robust uncertainty quantification.

Whether large medical foundation models will eventually supersede standard transfer learning paradigms for specific clinical applications remains an open question. Ultimately, resolving this debate will require the field to move beyond unidimensional accuracy metrics, adopting evaluation frameworks that rigorously align with the safety, interpretability, and reliability standards demanded by actual clinical workflows.

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